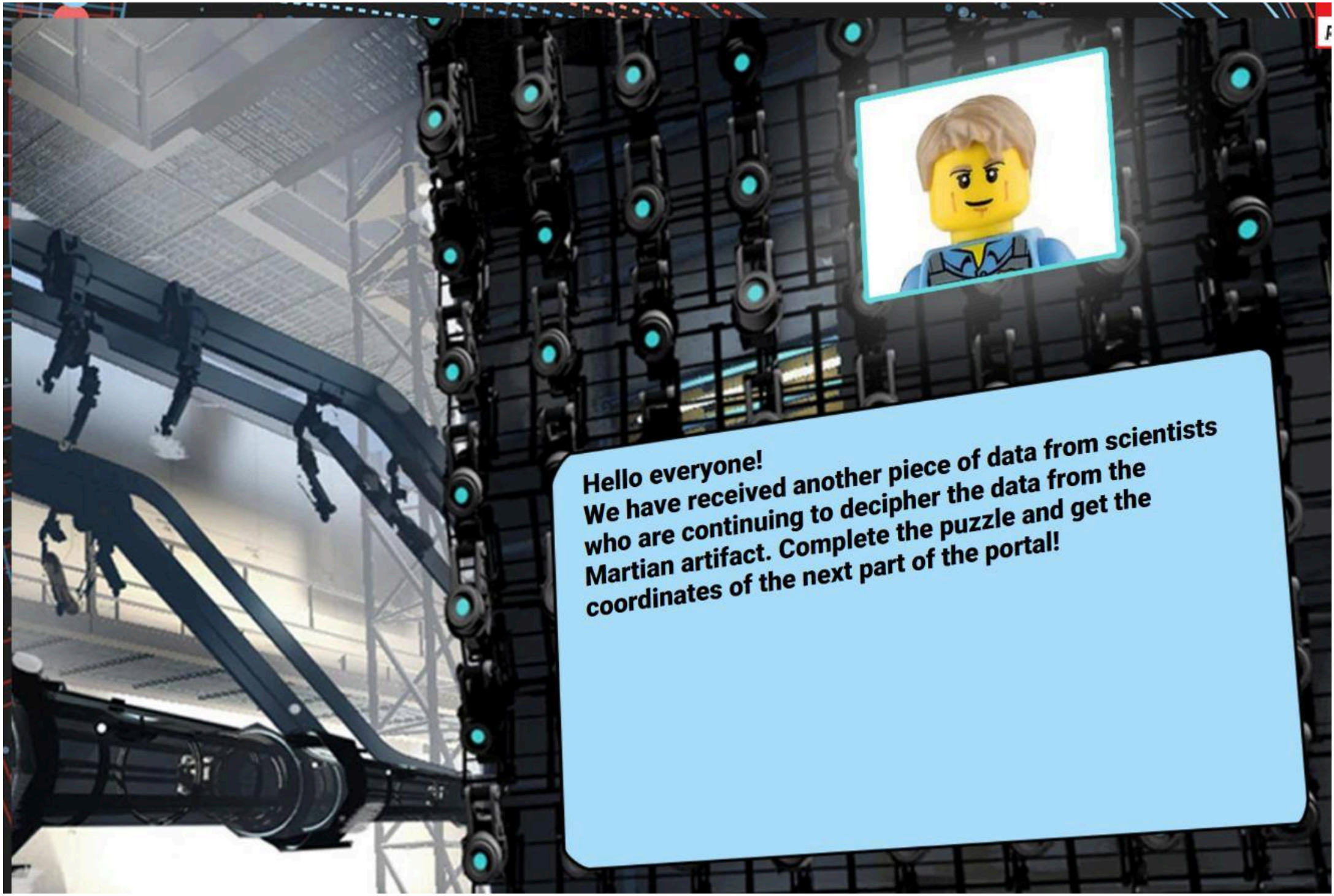


MazeRunner


Hello everyone!
We have received another piece of data from scientists who are continuing to decipher the data from the Martian artifact. Complete the puzzle and get the coordinates of the next part of the portal!



Did you recognize the country in which the portal element is hidden?

Egypt



The next part of the portal ended up in Egypt! This country has one of the oldest histories. Ancient Egypt is considered the cradle of civilization and one of the first centers of early writing.



Discuss!



What do you know about Egypt?
Where do you think the portal parts might be hidden?

Egypt



Egypt has a lot of historical monuments. The world-famous Pyramids of Giza, the Sphinx, the Valley of the Kings, Thebes, and the Karnak Temple are of constant interest to historians and tourists.



Egypt



Egypt has a lot to offer tourists. This includes rest on the Red or Mediterranean seas, and excursions to the monuments of ancient civilizations.



TASK

TASK

- create a robot to search for objects in pyramid mazes

Steps

- learn about the structure of the pyramids
- build and program a robot capable of moving through mazes
- search and collect parts of the portal

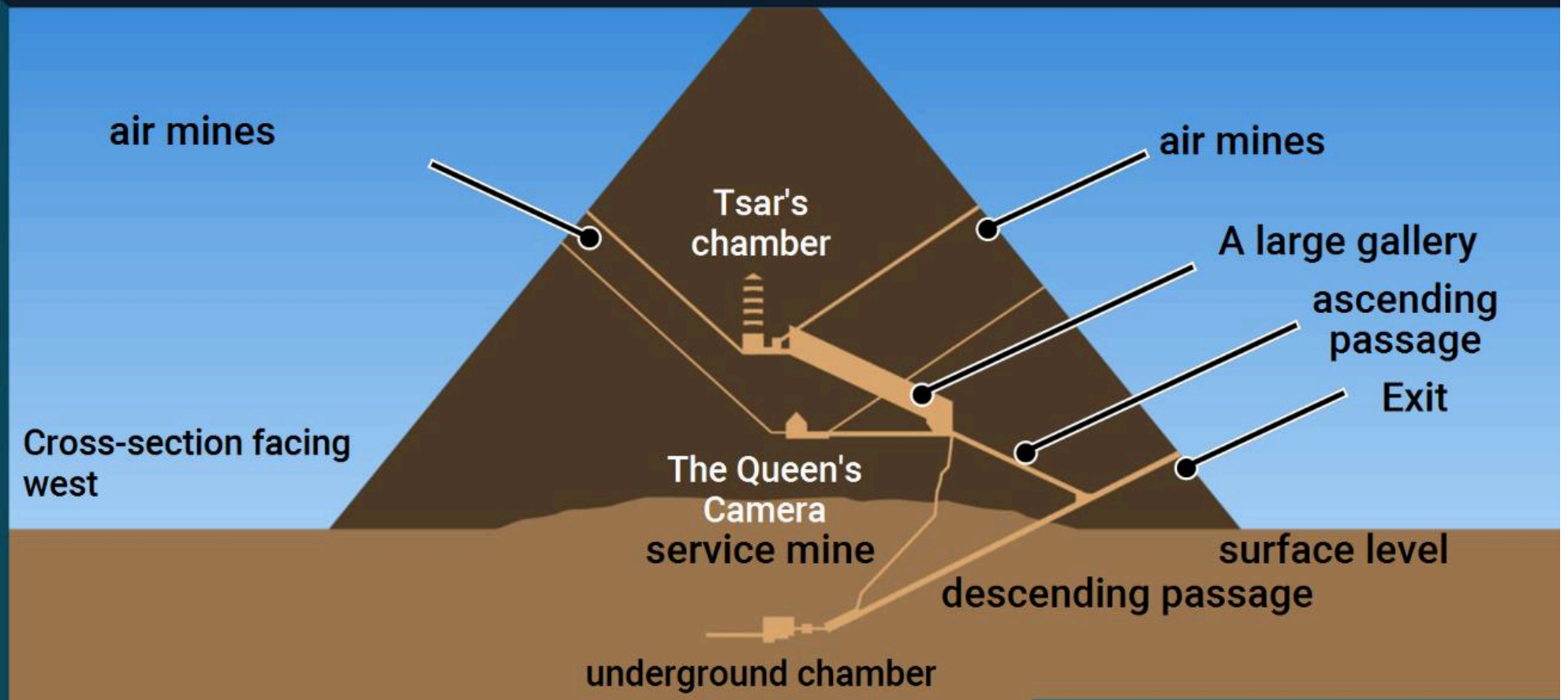


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The structure of the Cheops` pyramid



Inside, the pyramid has a complex system of shafts and passages connecting various ritual halls. Scientists suspect that the currently known structure of the pyramid is not complete, and it may have more than one secret hall.



Meet MazeRunner!



This is a robot perfectly adapted to move through the narrow corridors of the pyramid. It is very maneuverable and has a sensor to determine the distance in front of it. And the manipulator will allow you to bring the found objects!

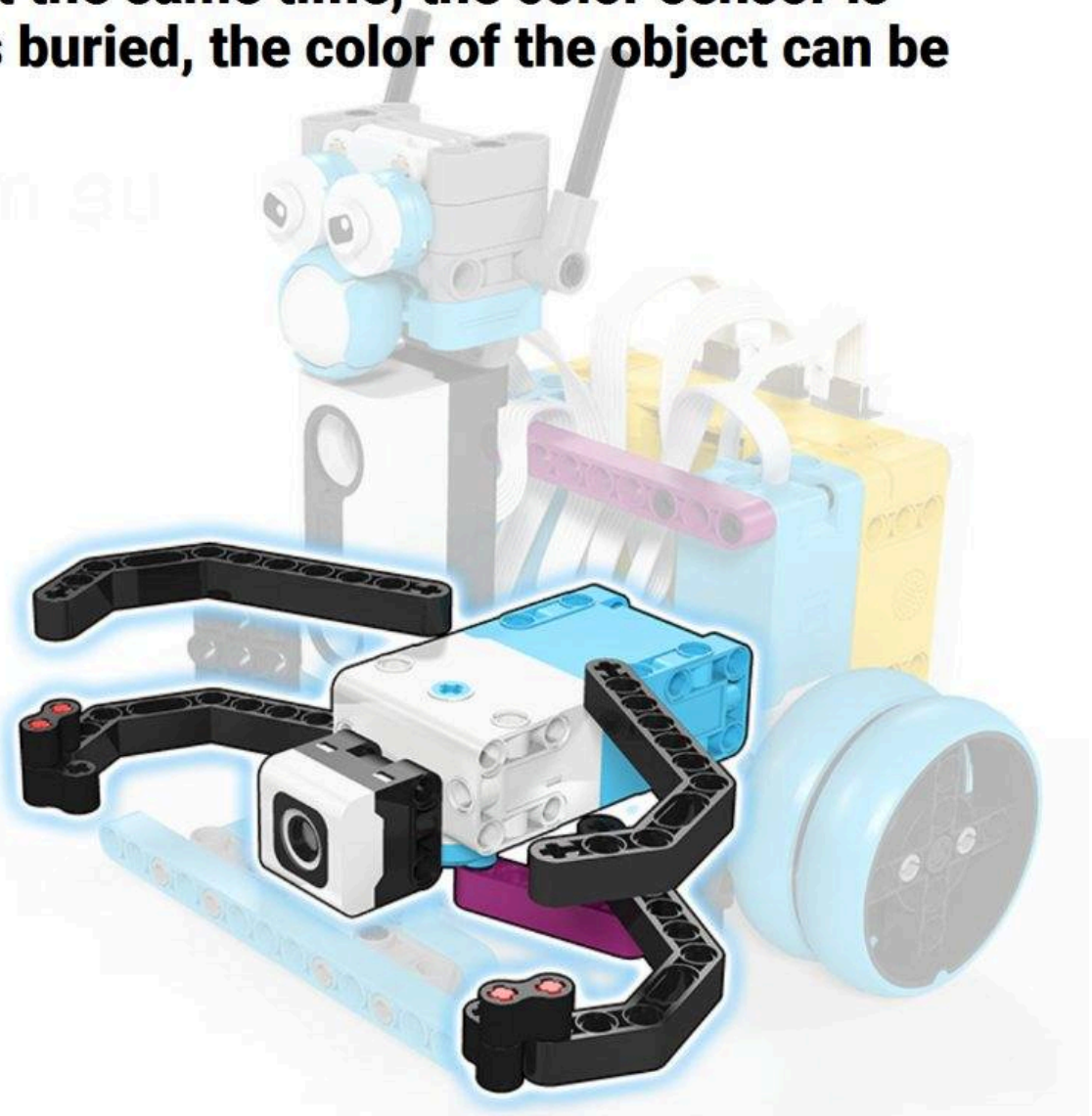


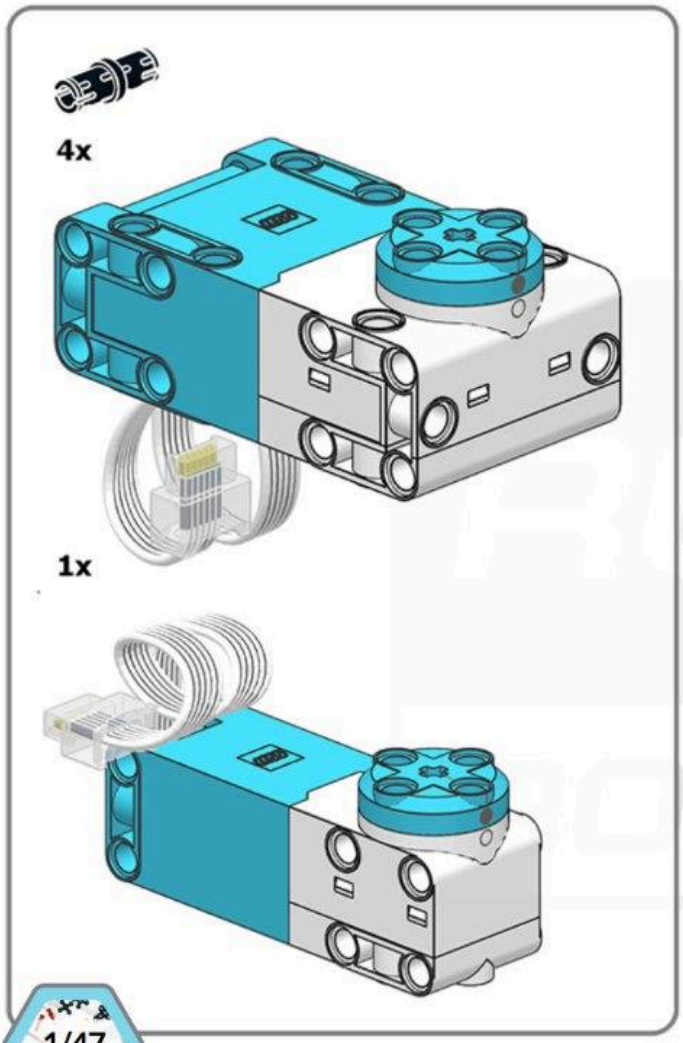
Design features

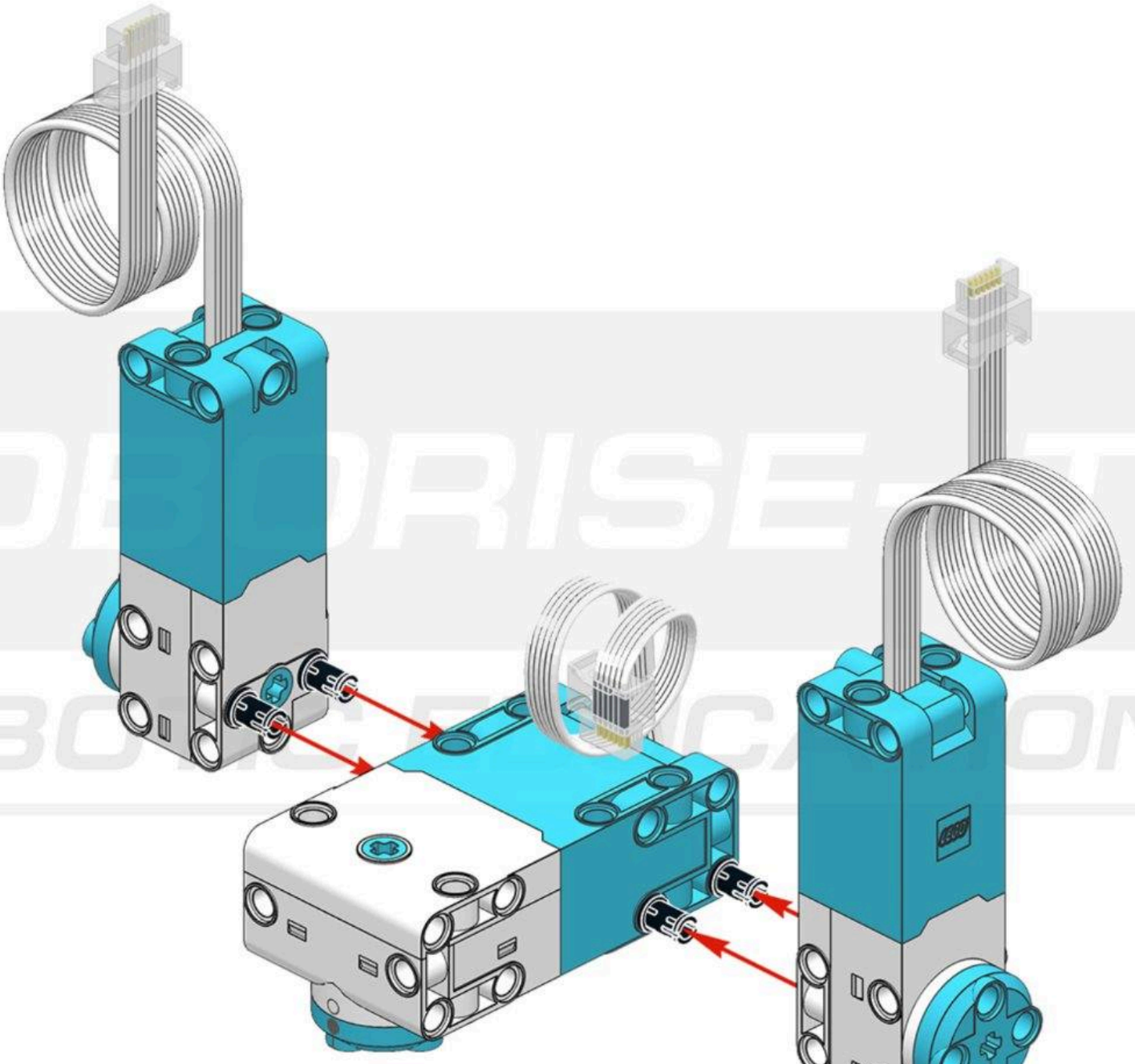
The lower part of the claws of the manipulator has rubber parts that reduce the slipping of objects. At the same time, the color sensor is placed so that when the object is buried, the color of the object can be accurately determined

Find:

- **Large motor**
- **Claws and rubber parts**
- **Color sensor**



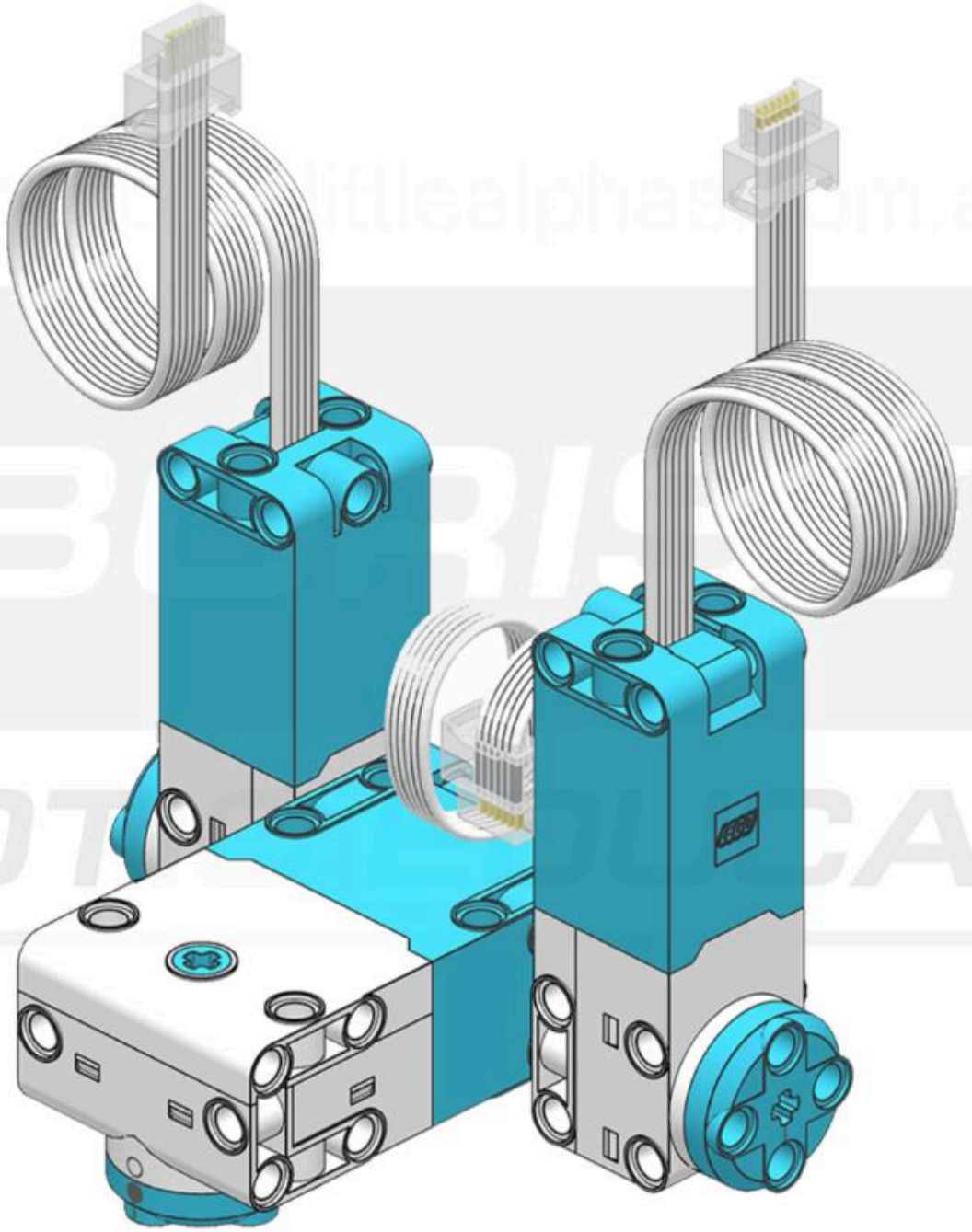




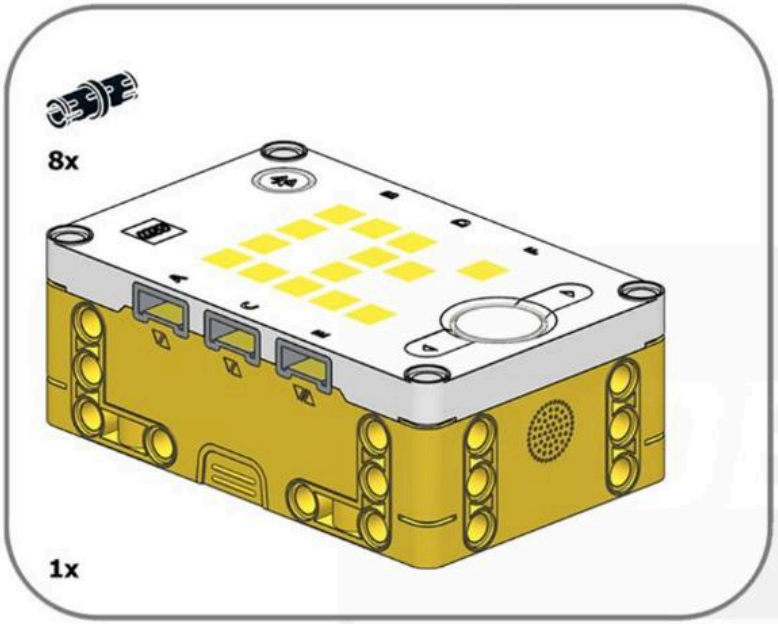
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Model by Ihnat Khliebniko

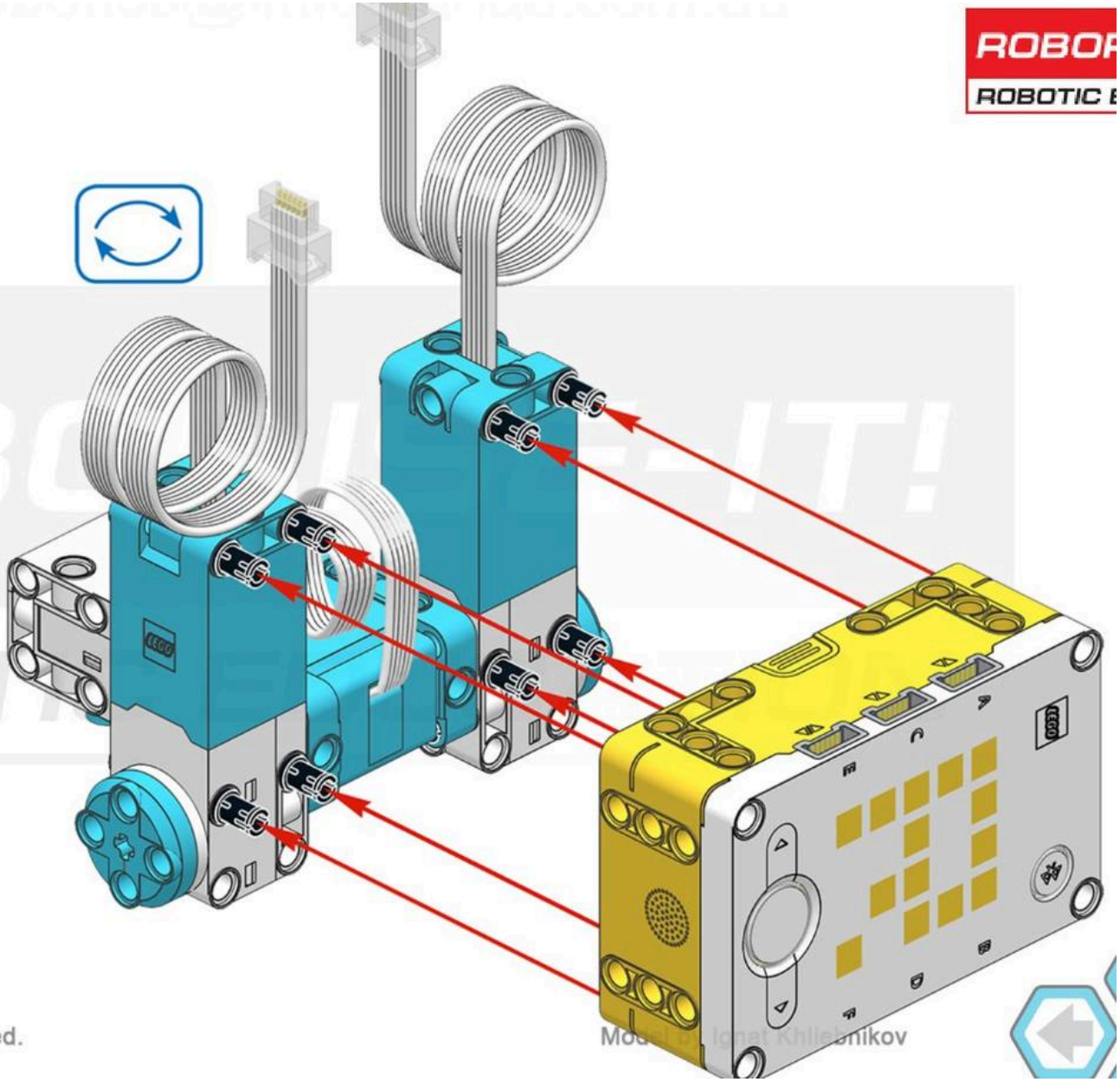
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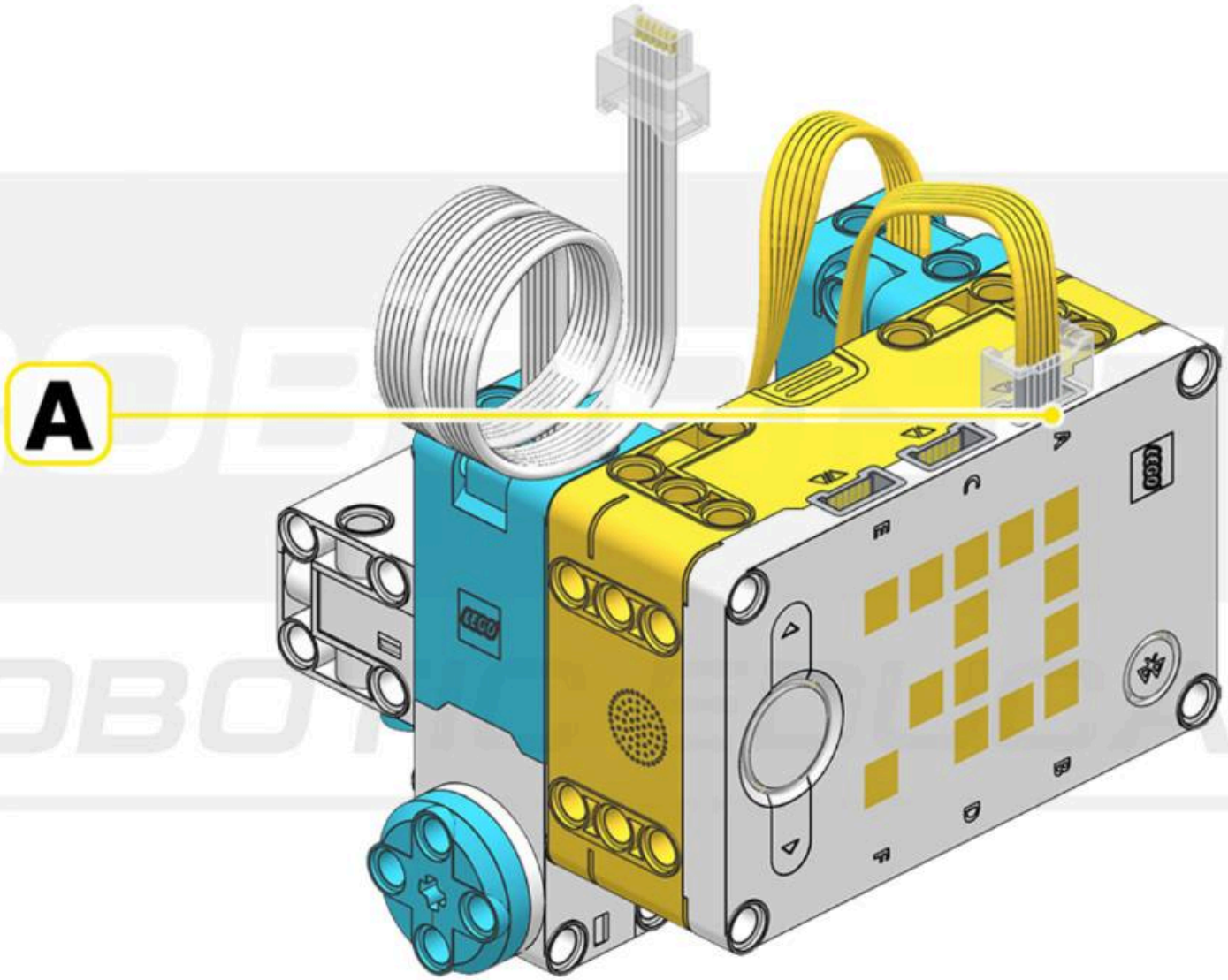


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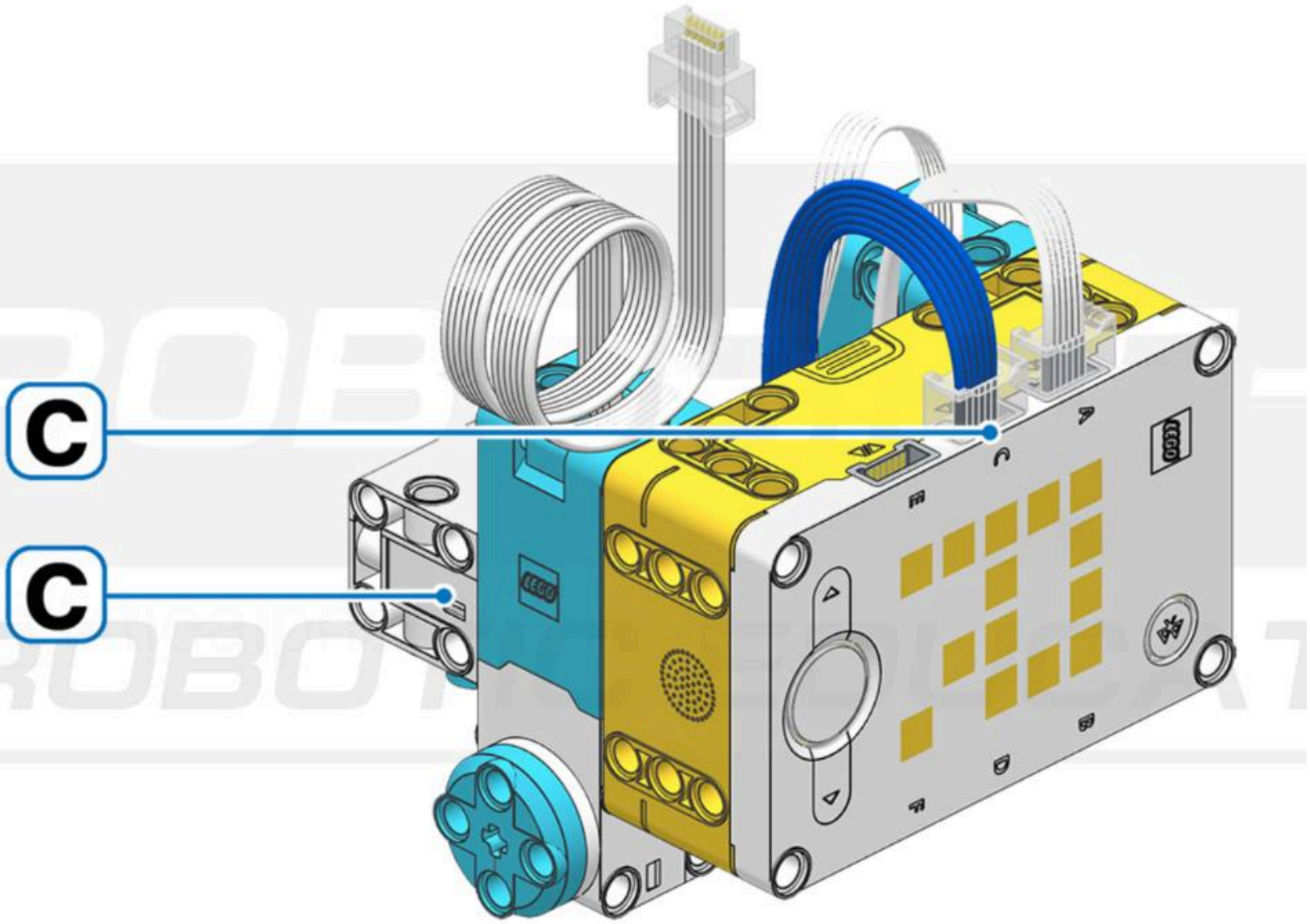


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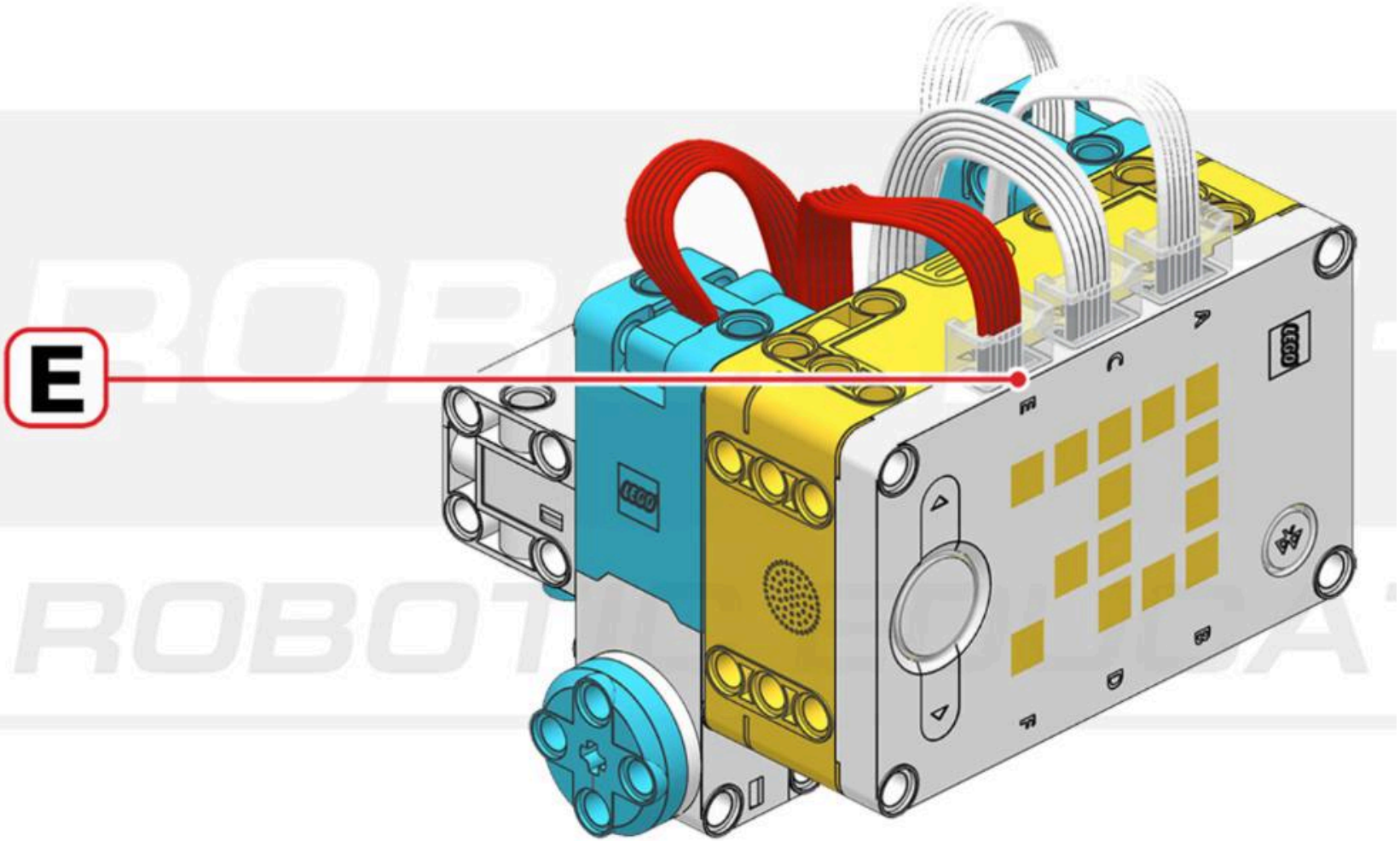
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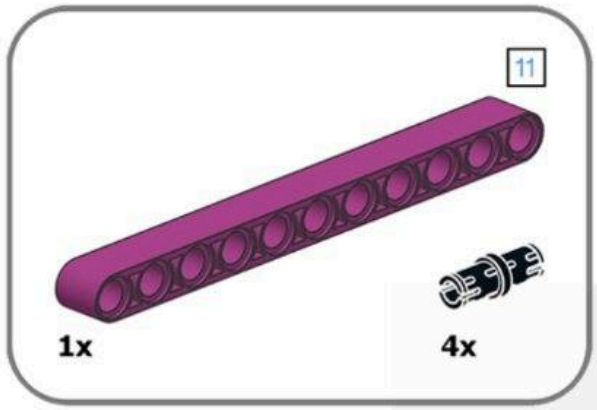


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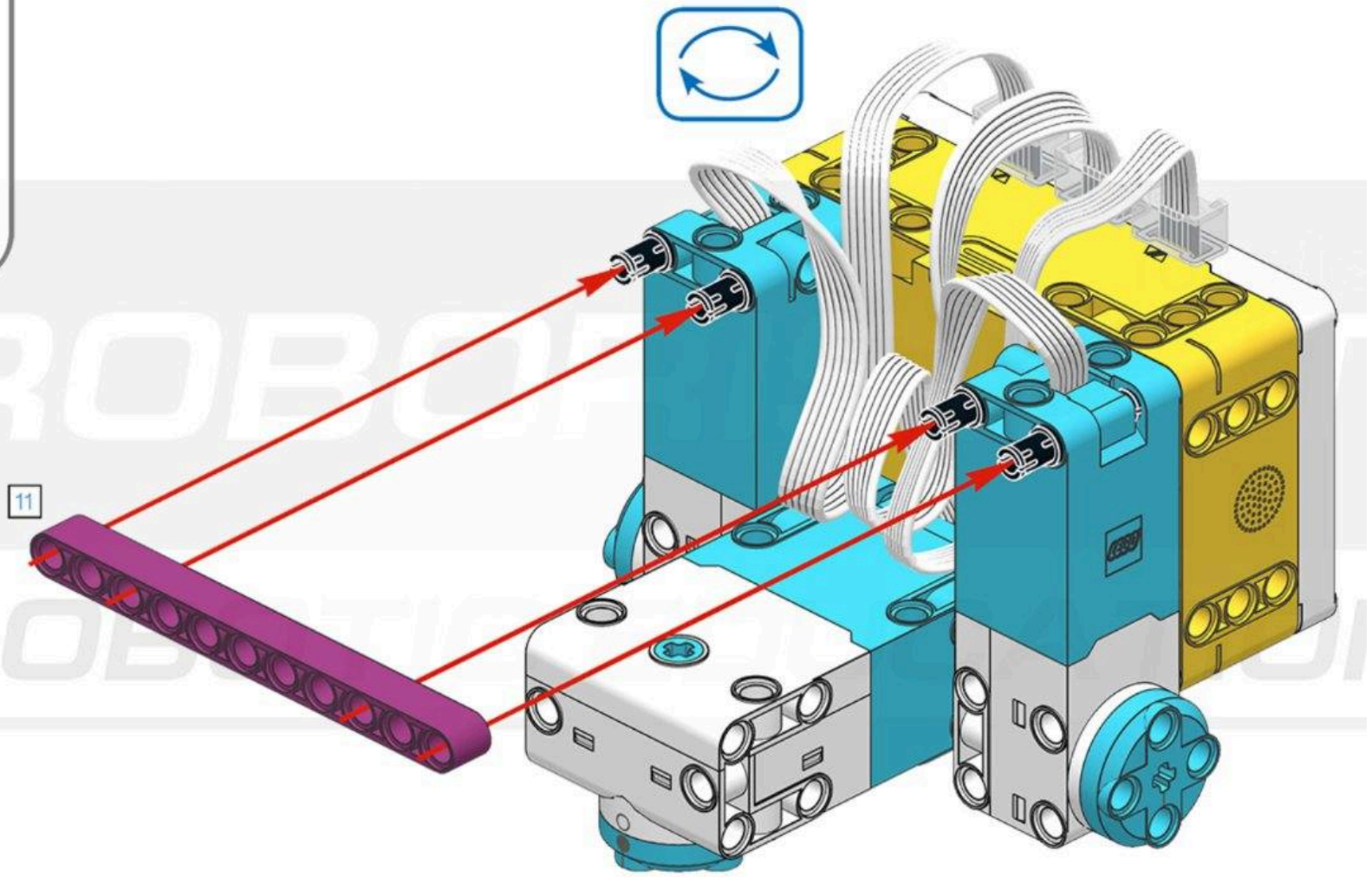


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8





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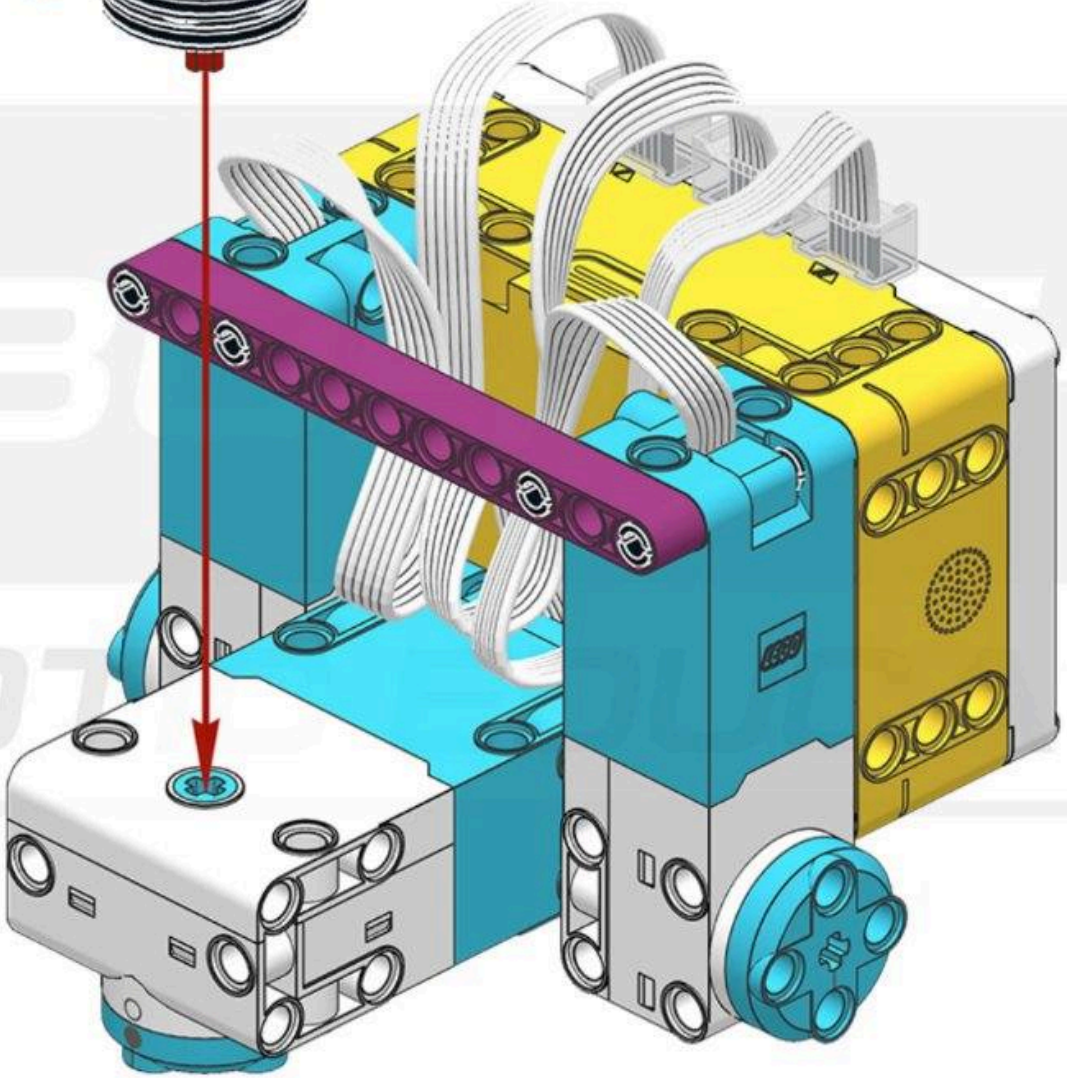






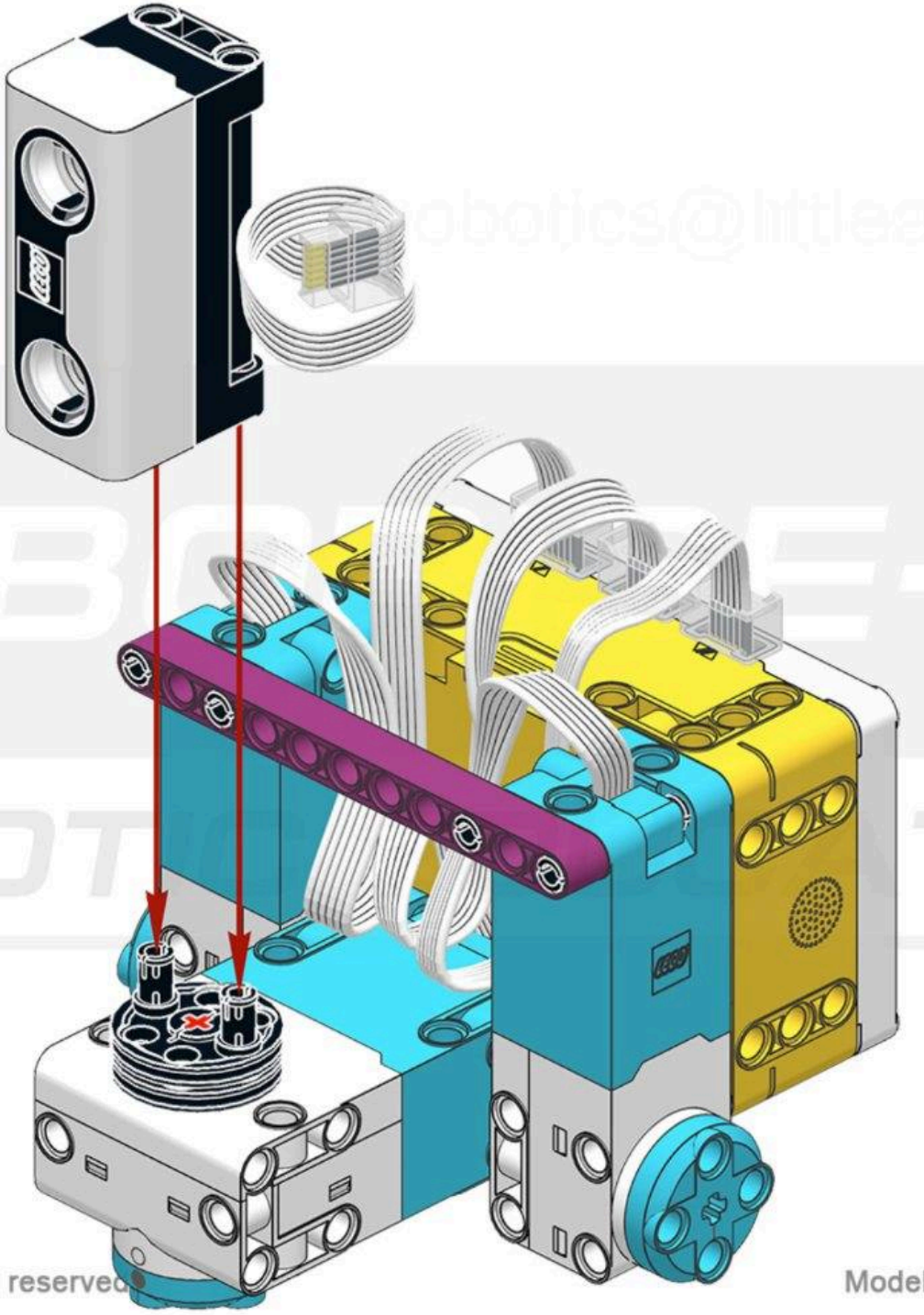






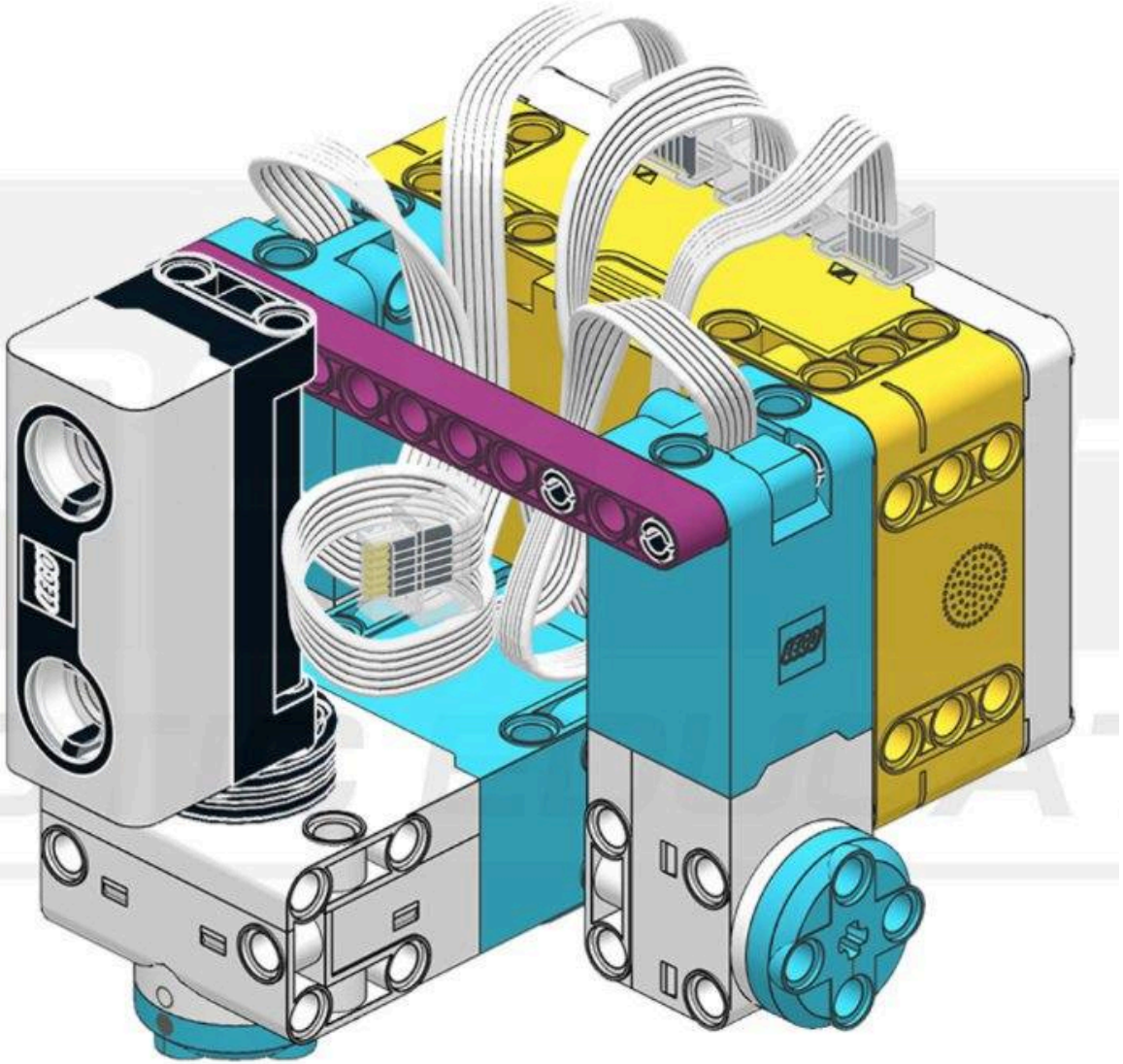


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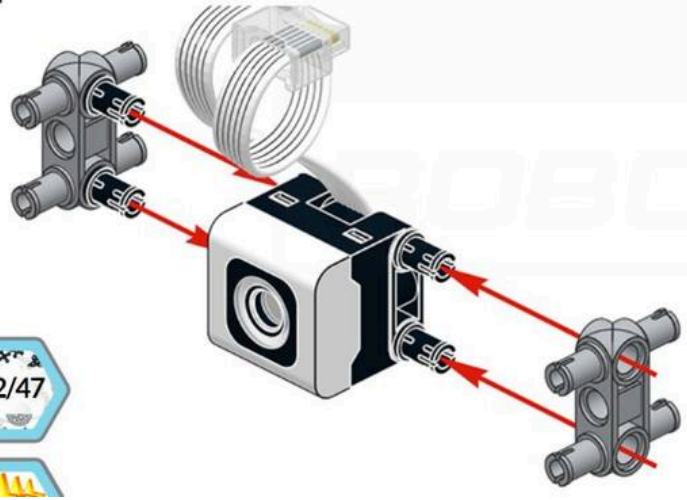
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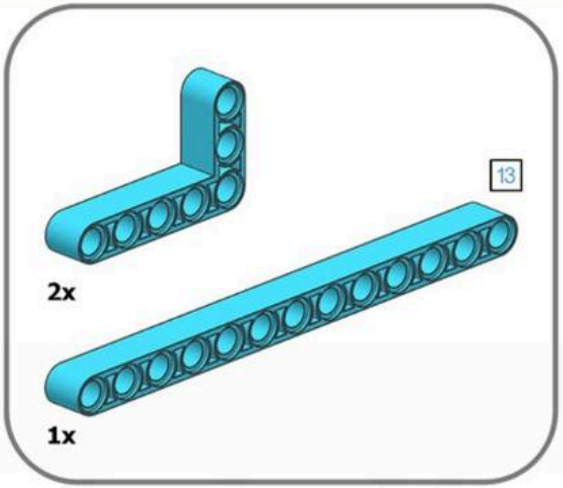
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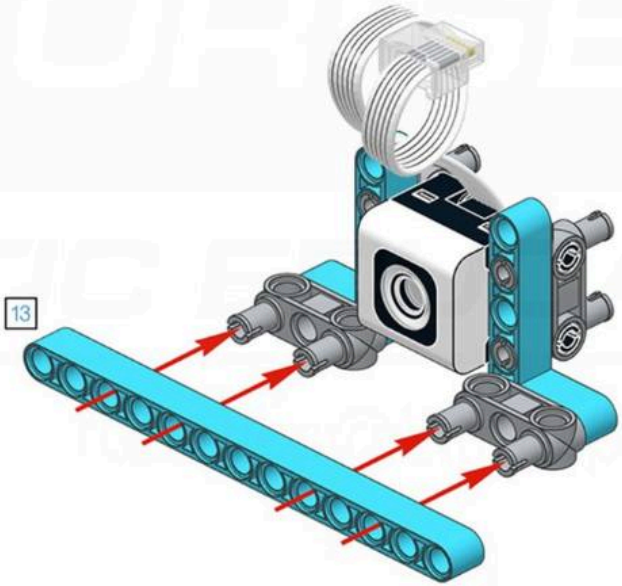


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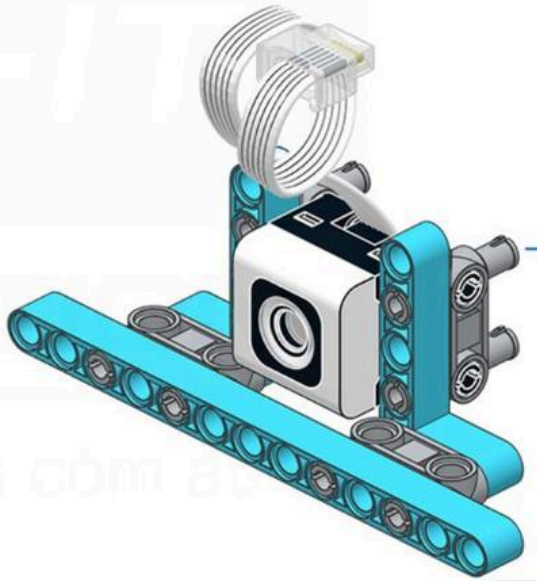


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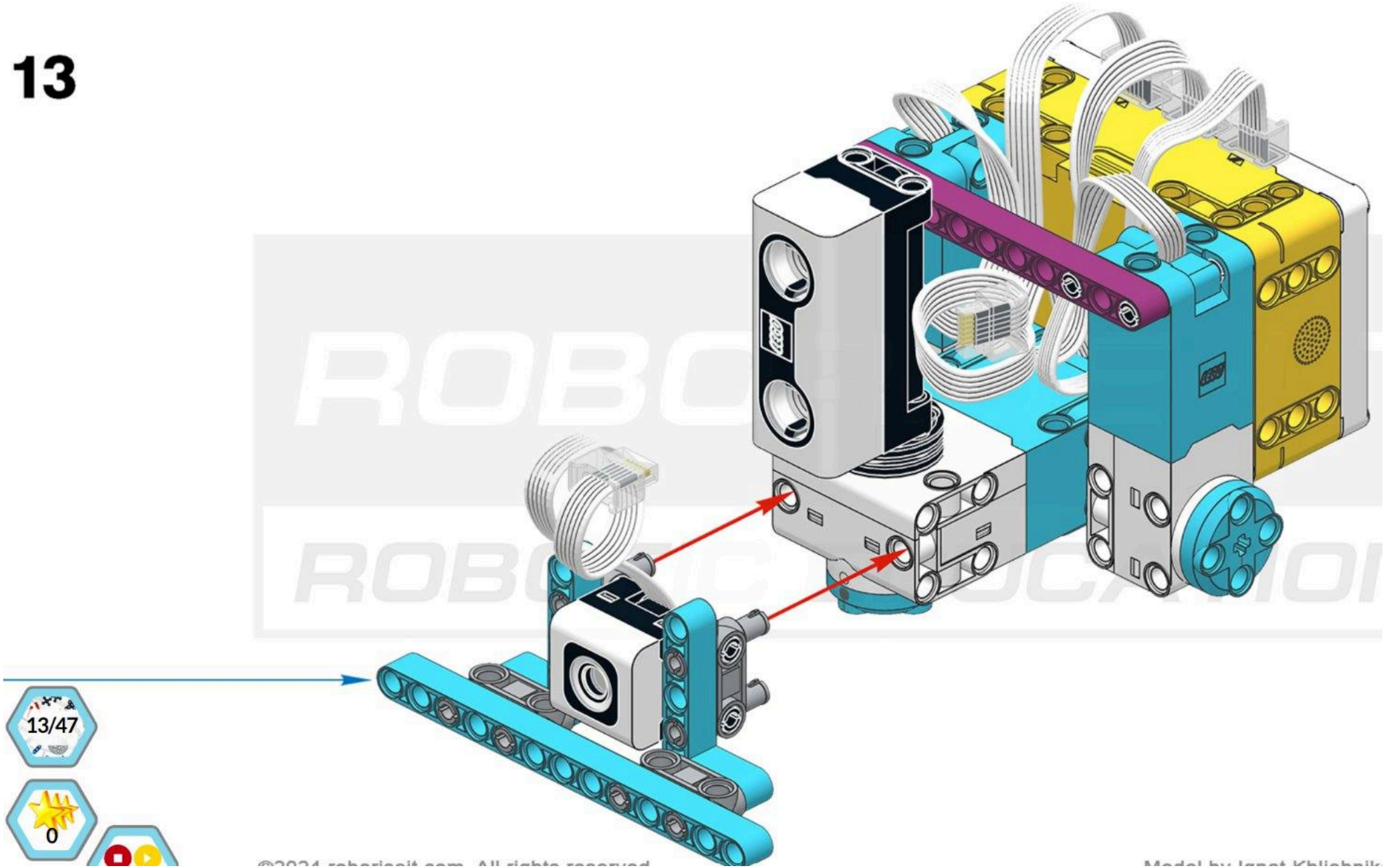


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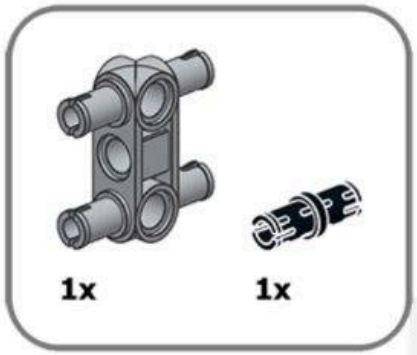


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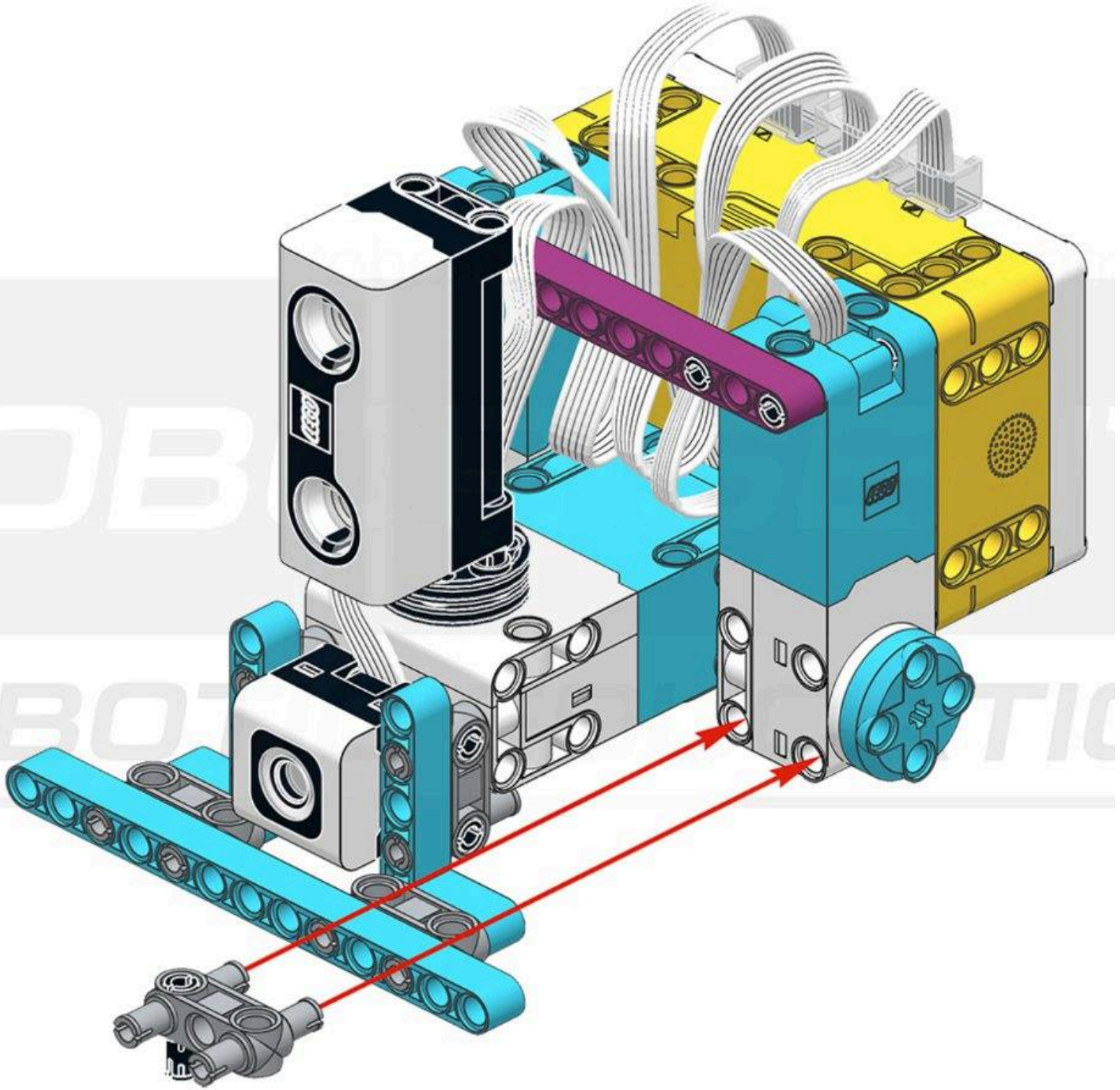


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Model by Ignat Khlichnik

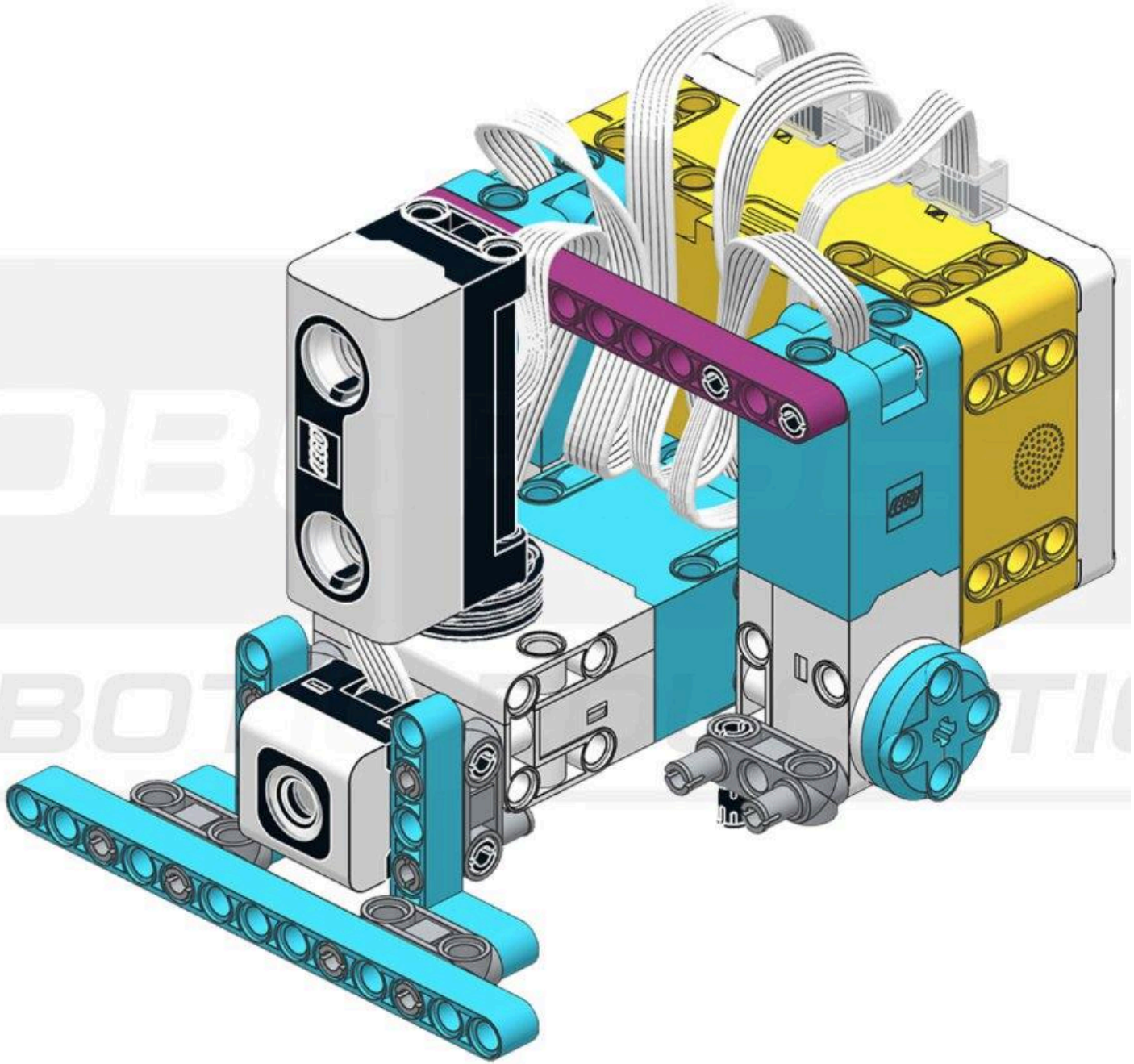


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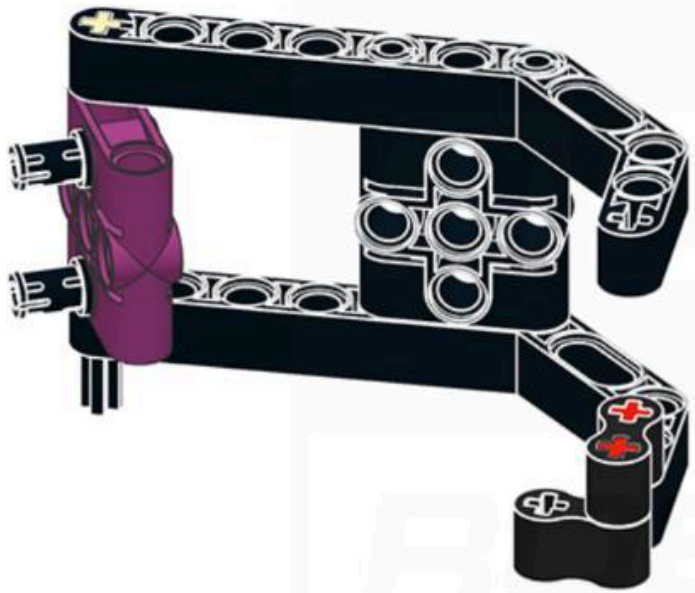


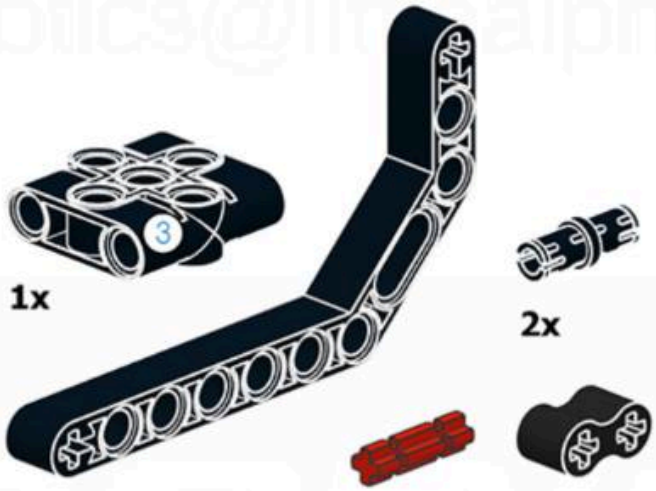
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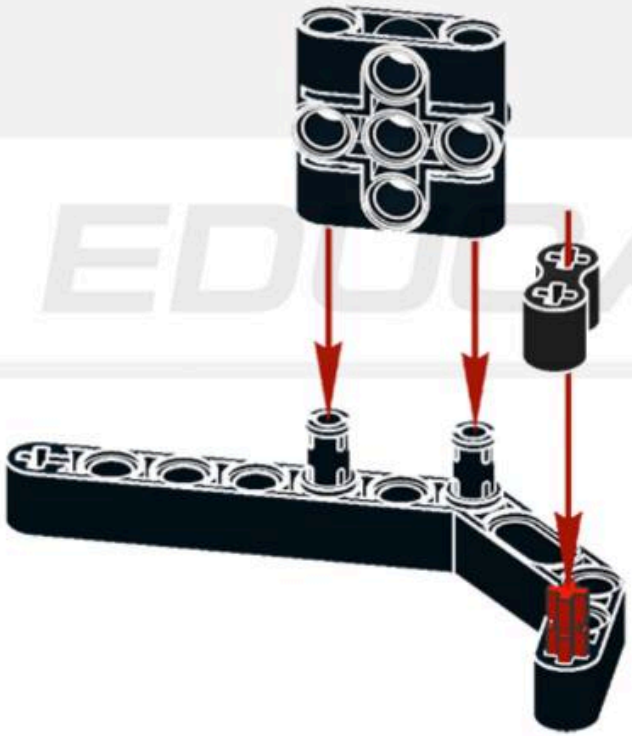


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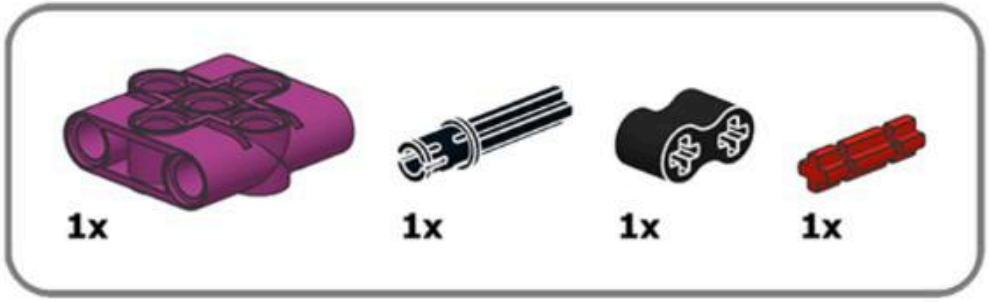




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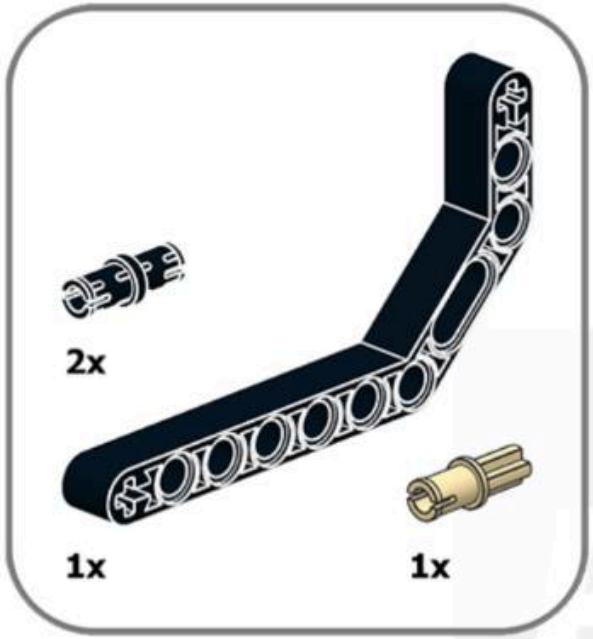


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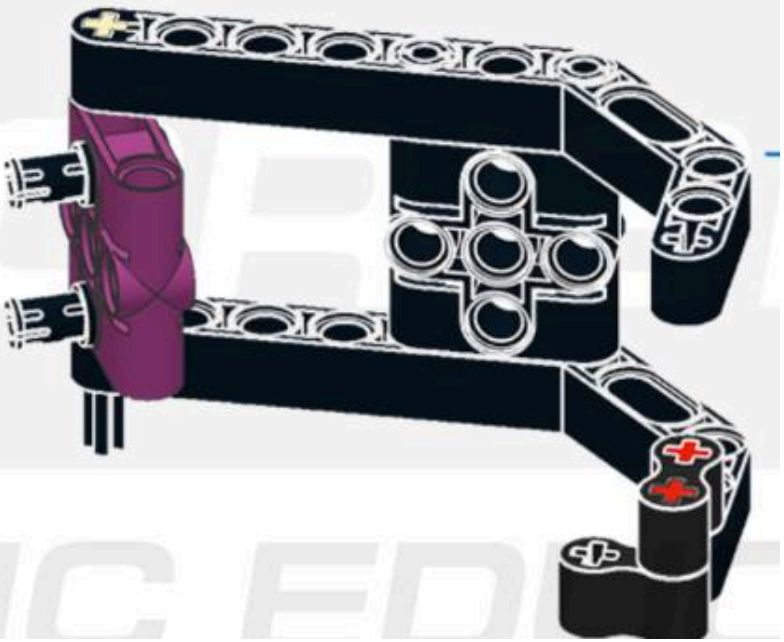
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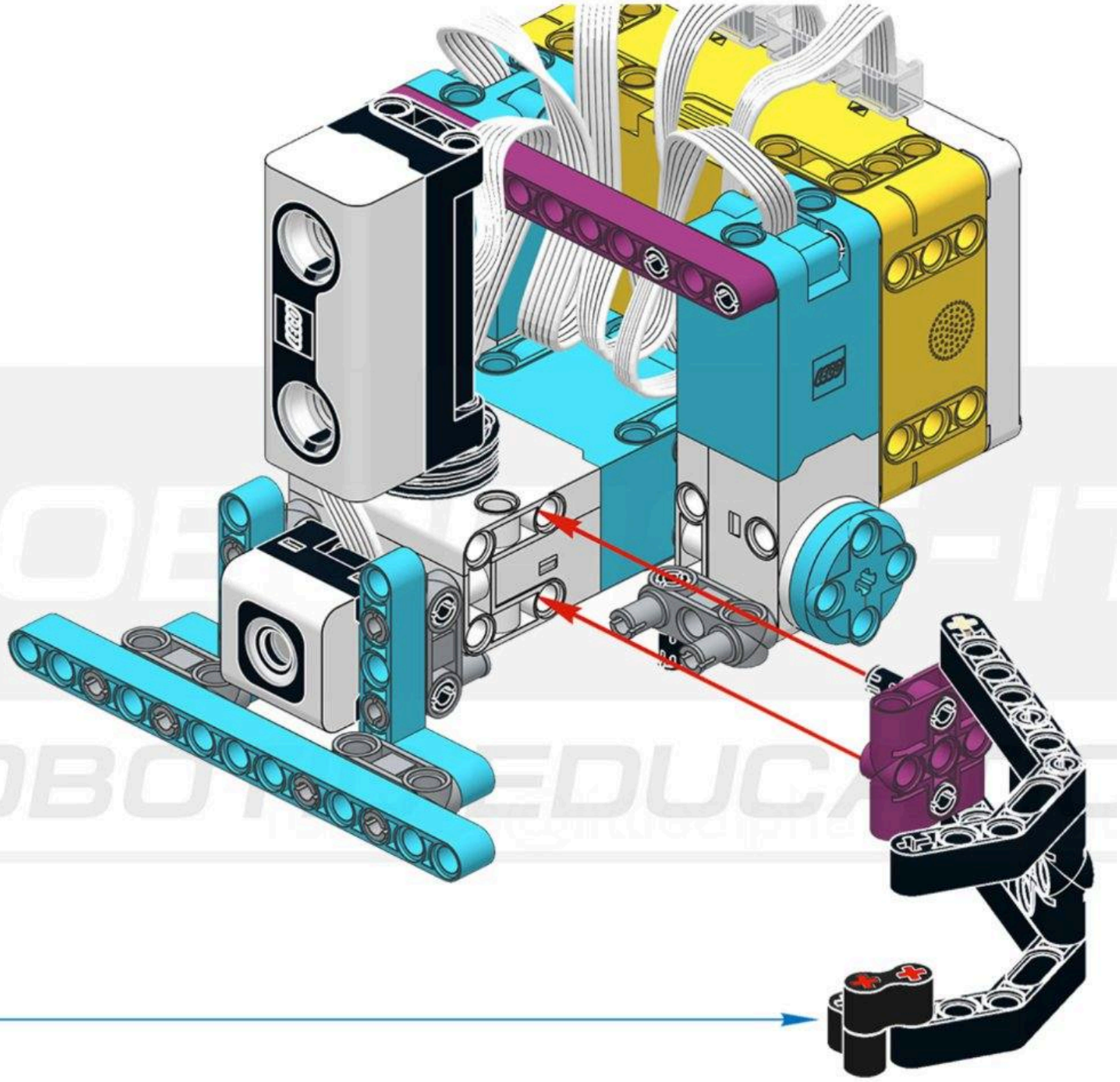


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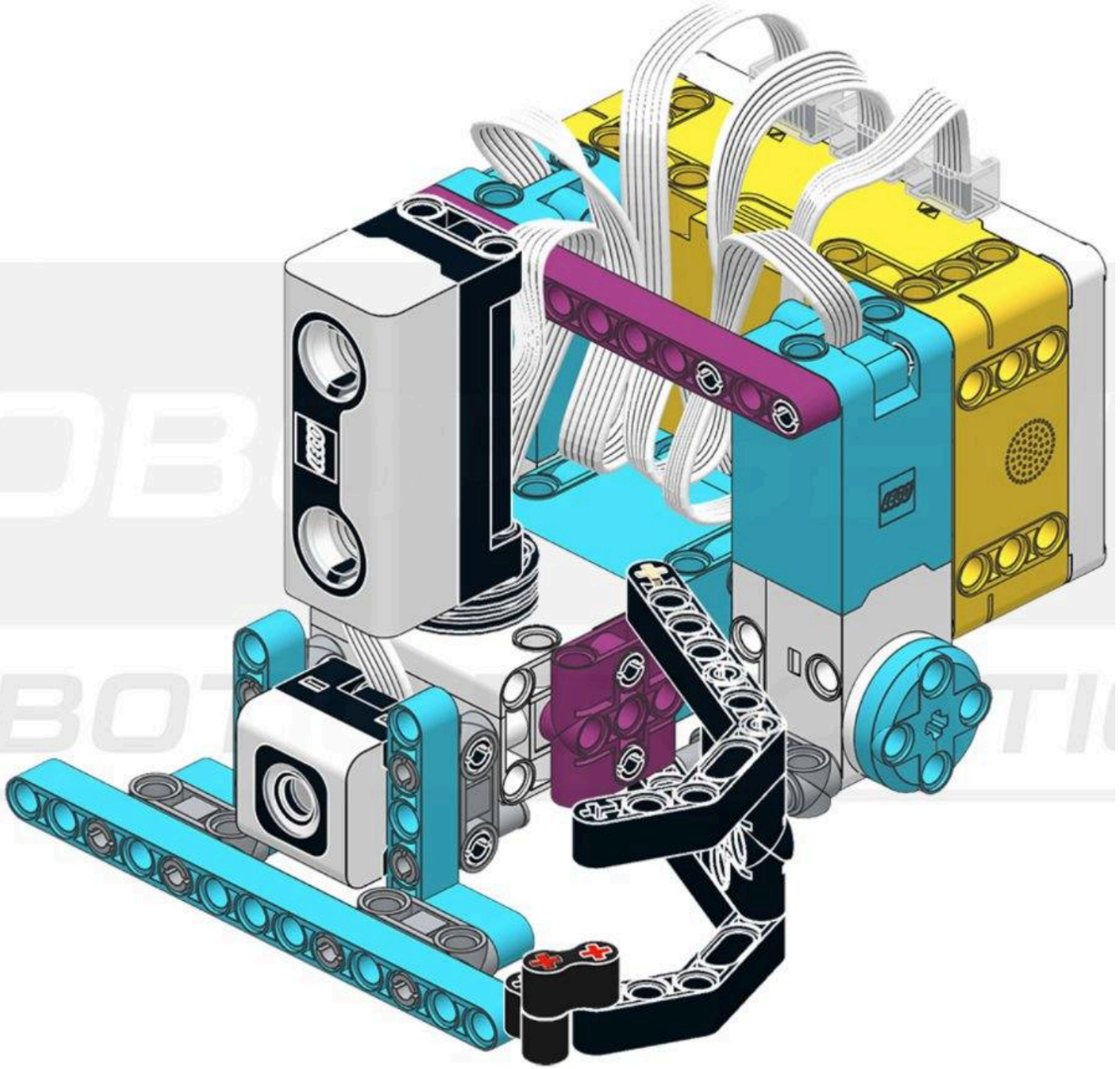
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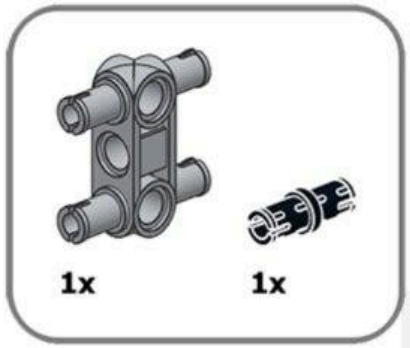


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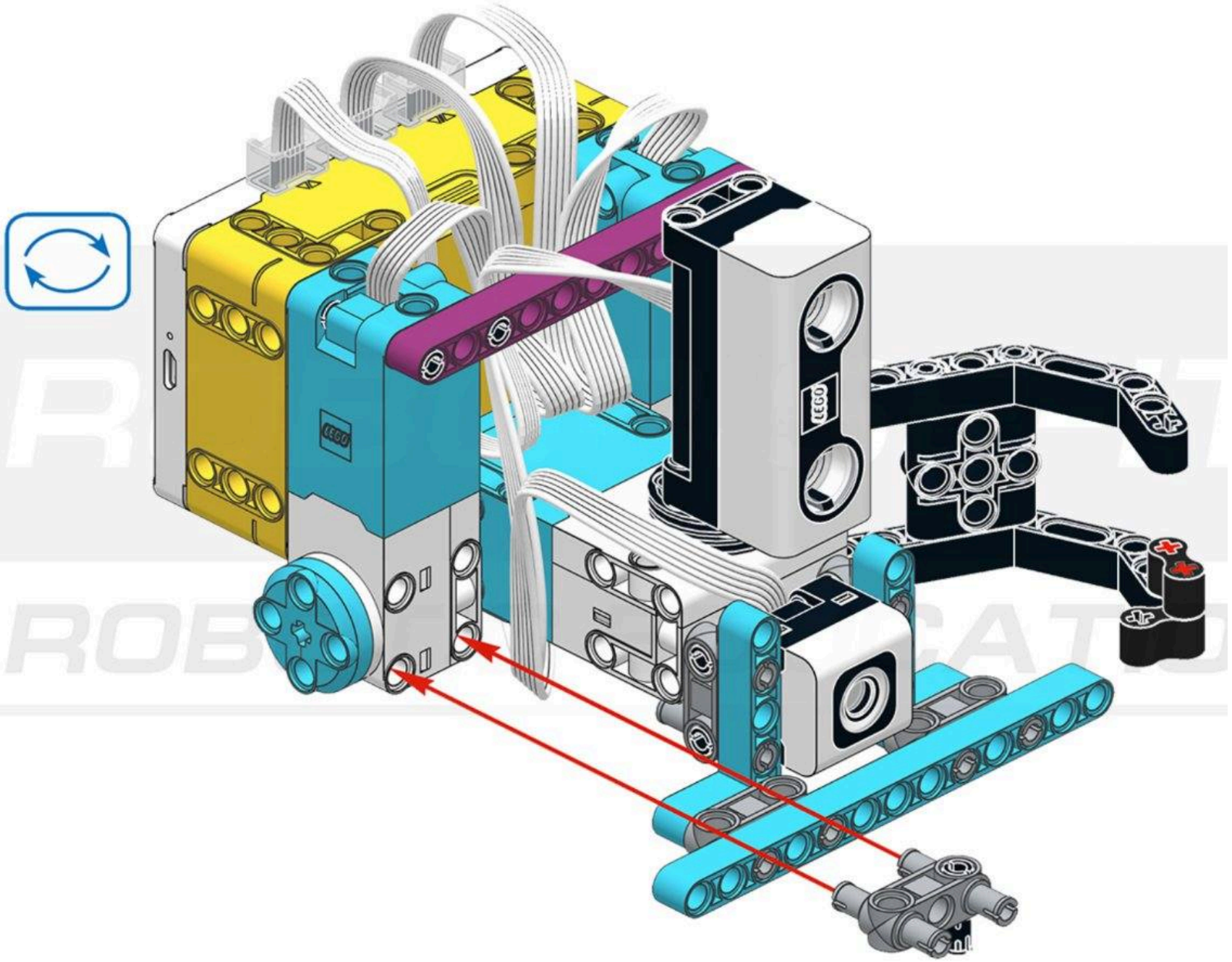


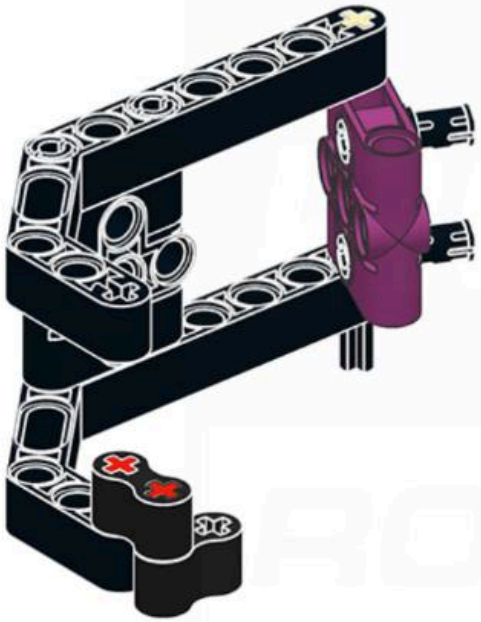
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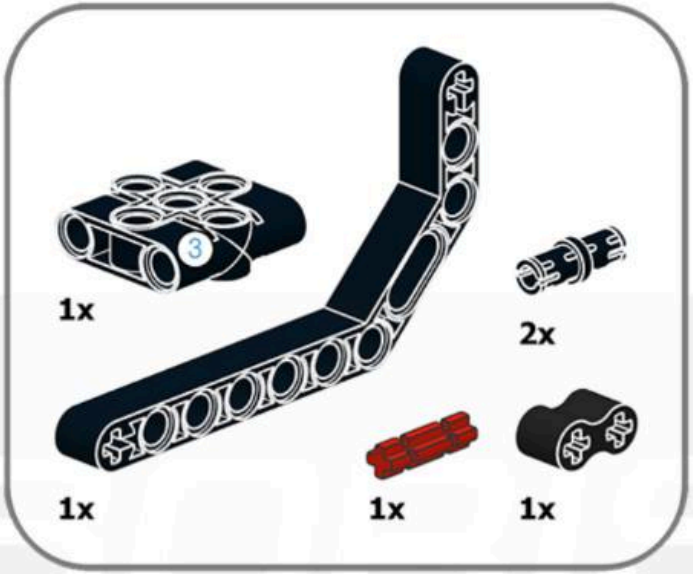




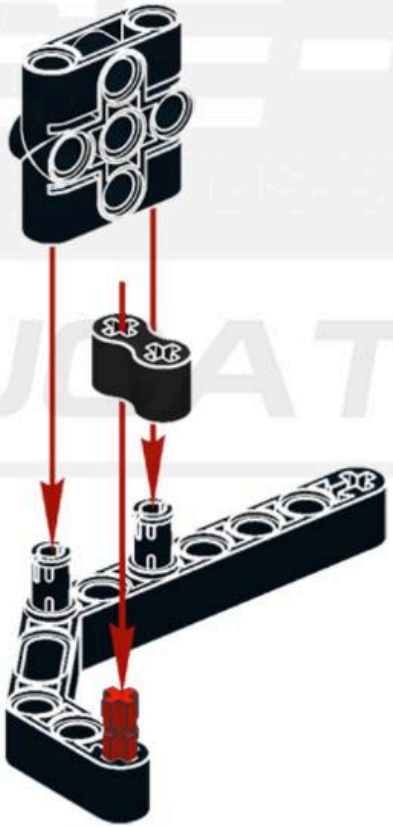
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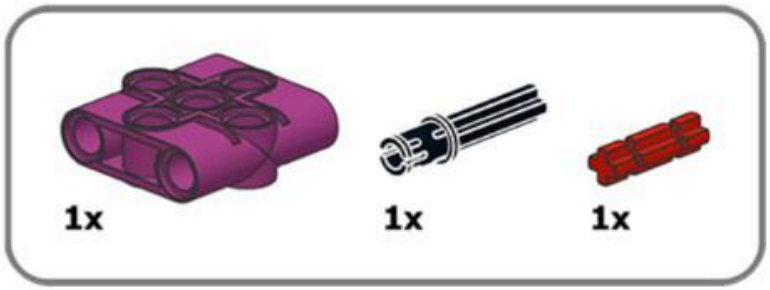




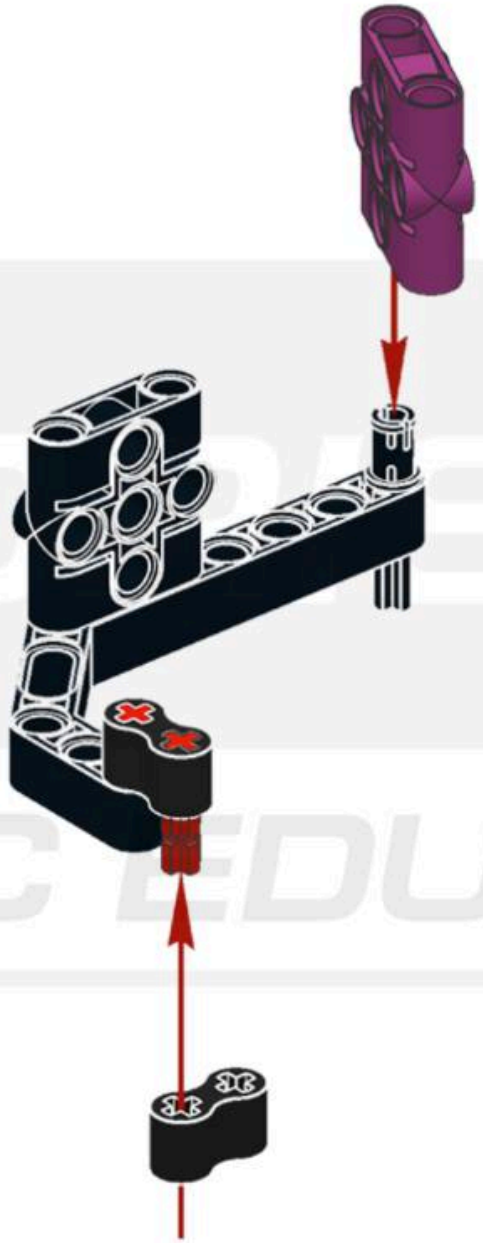
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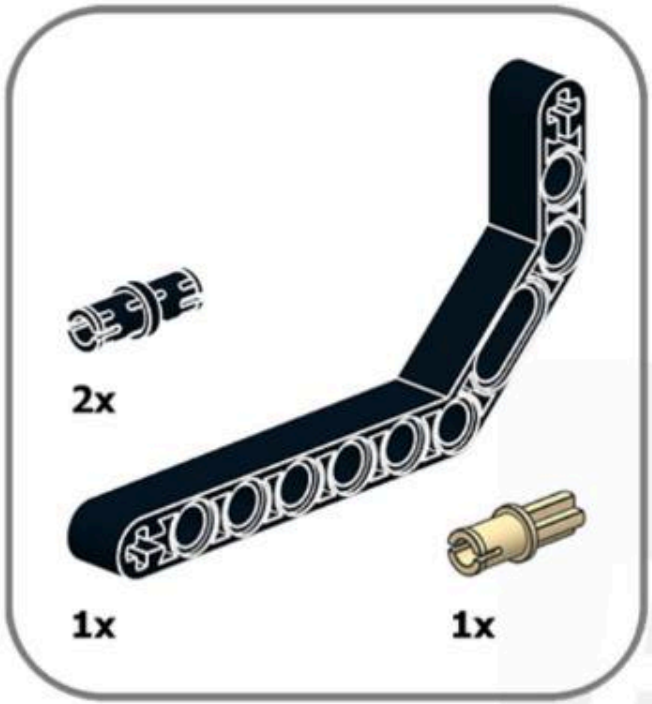
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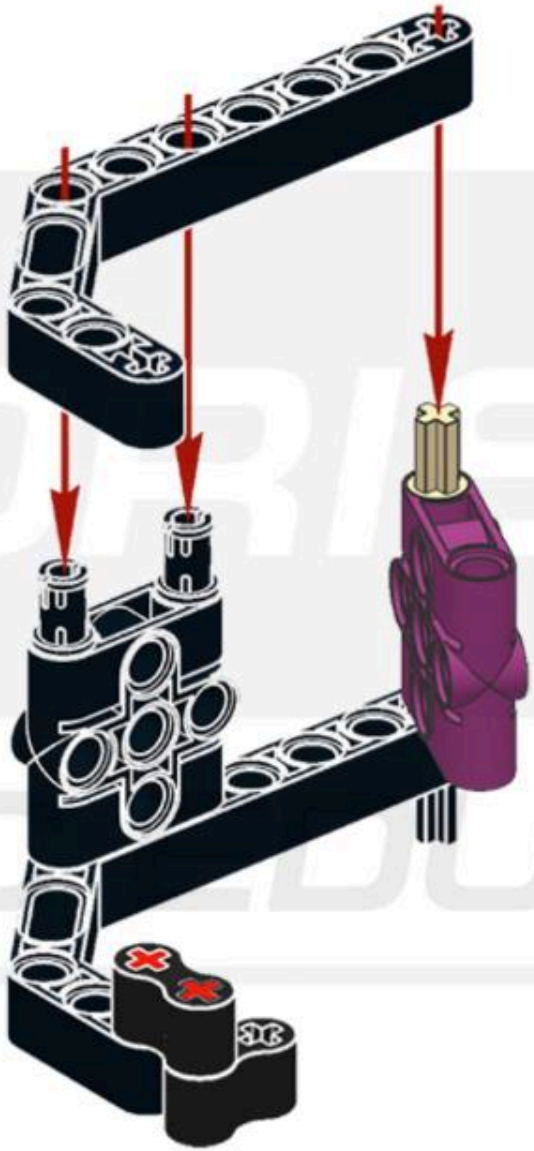
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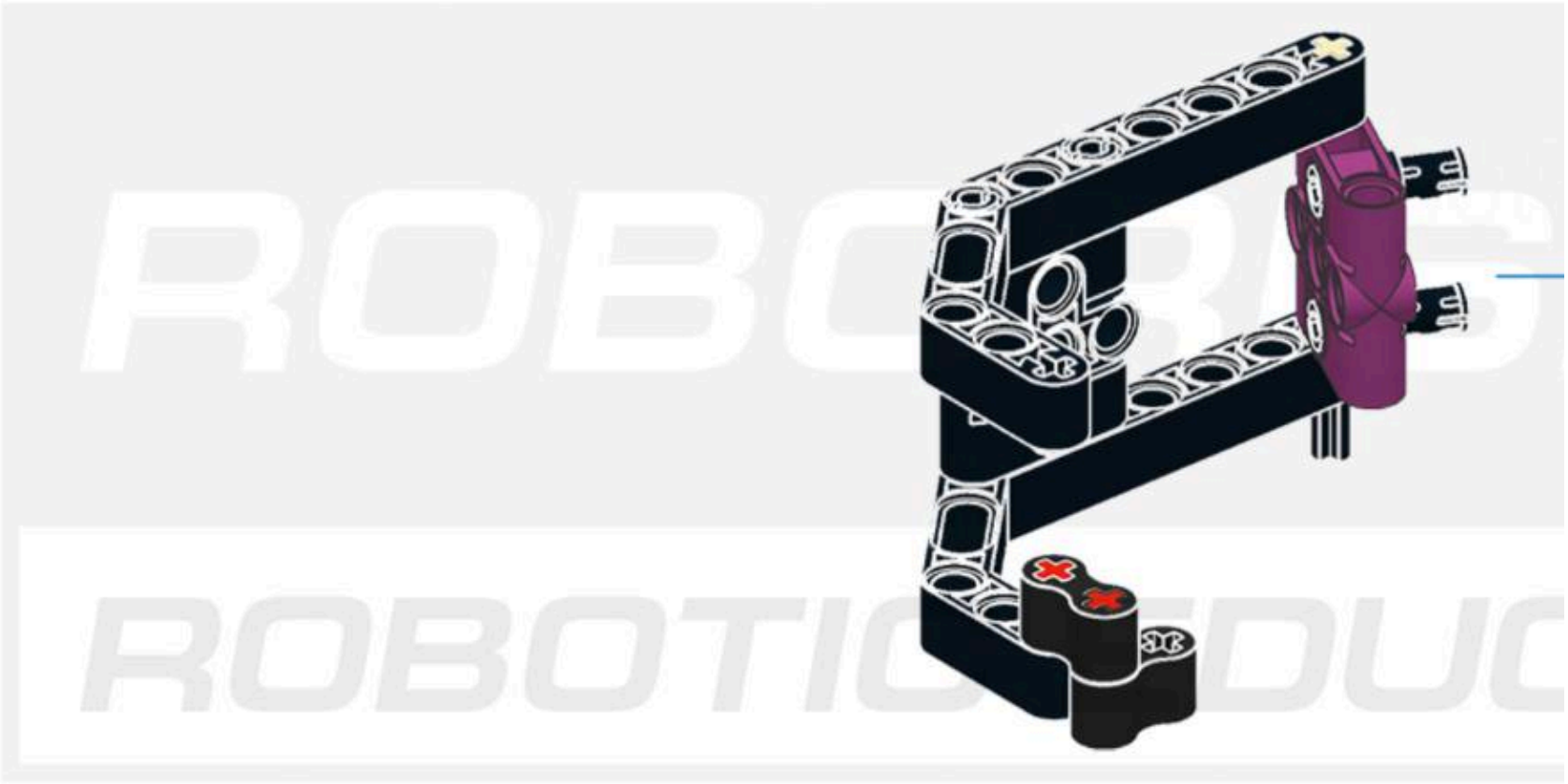


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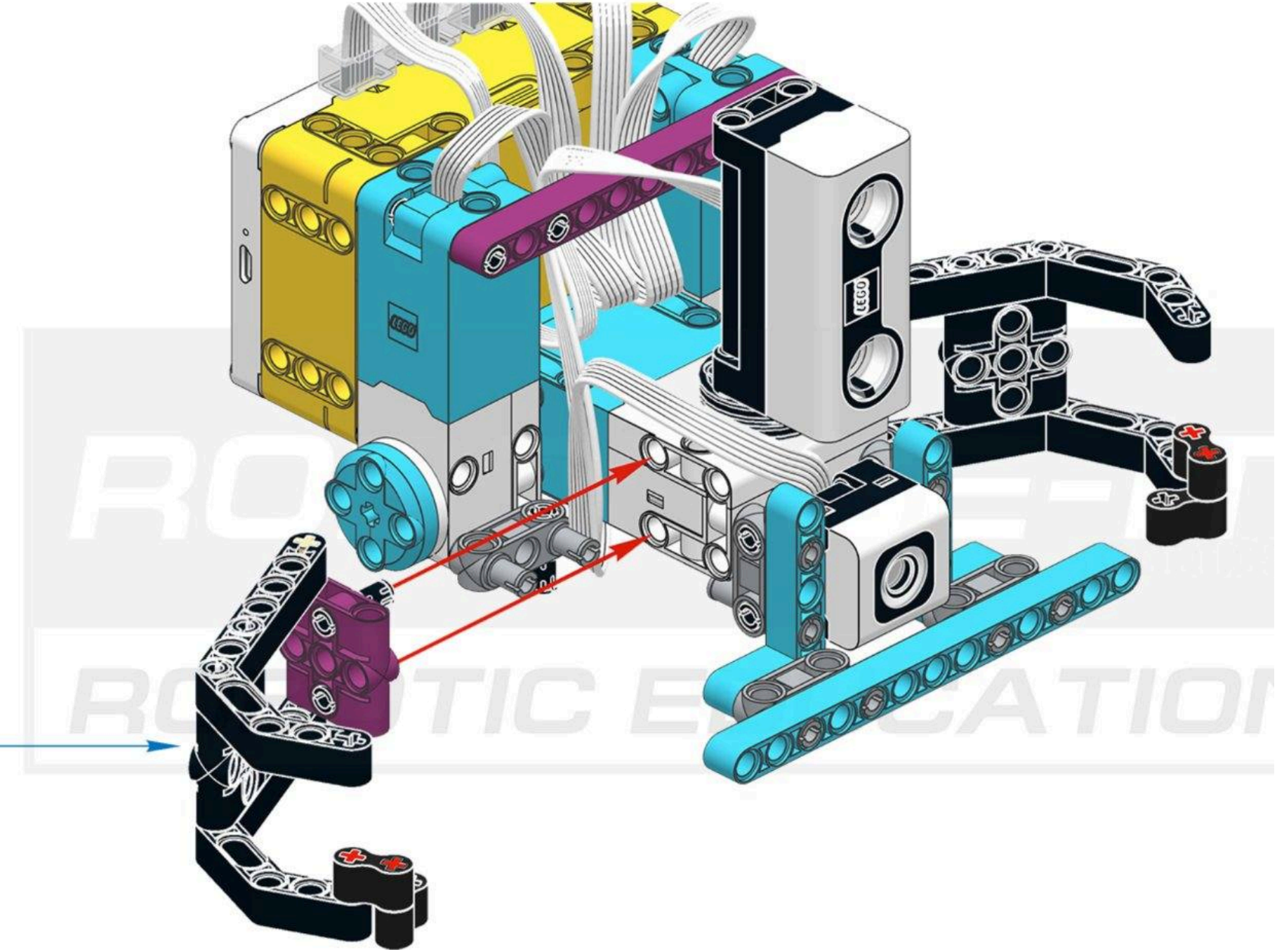




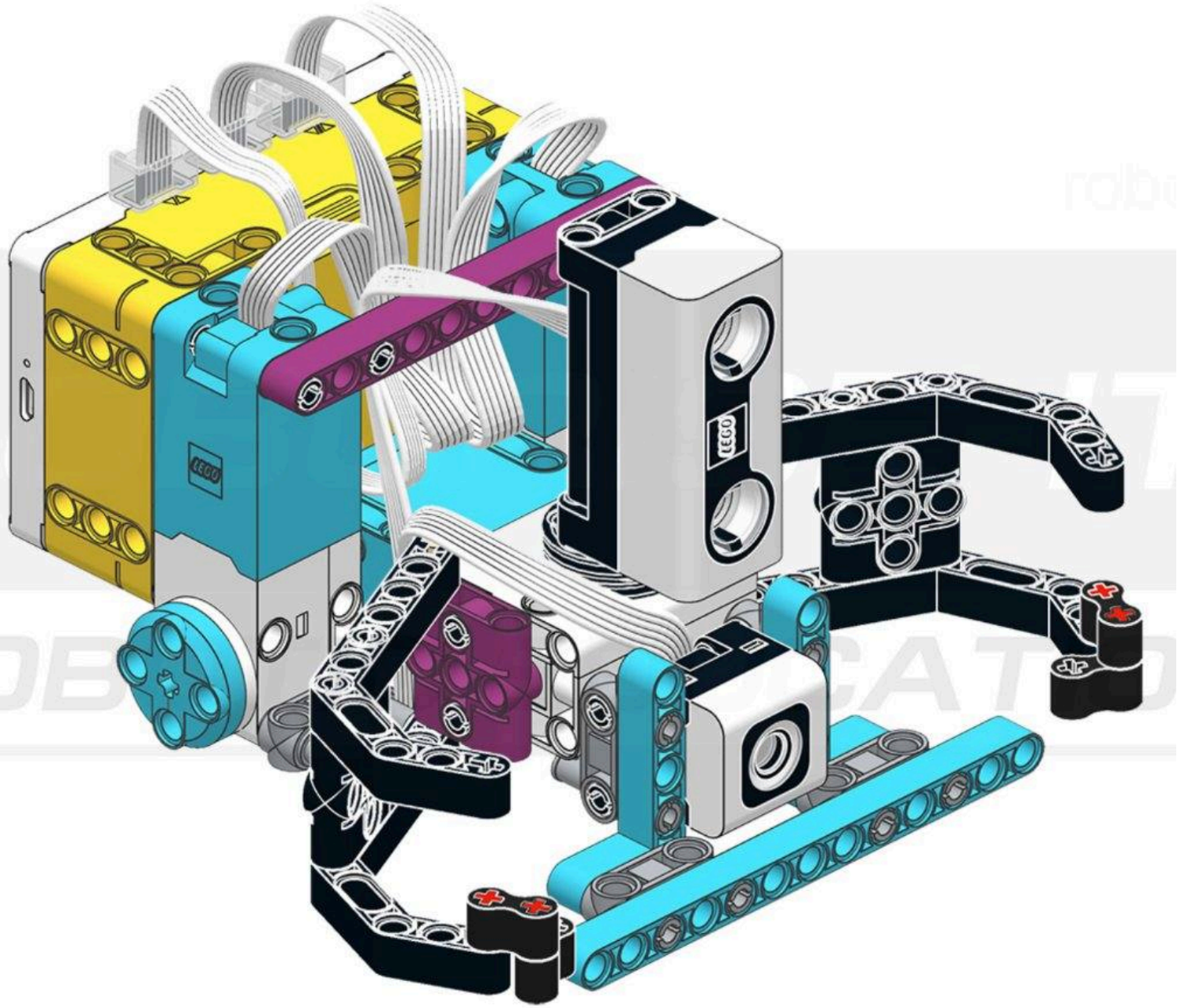
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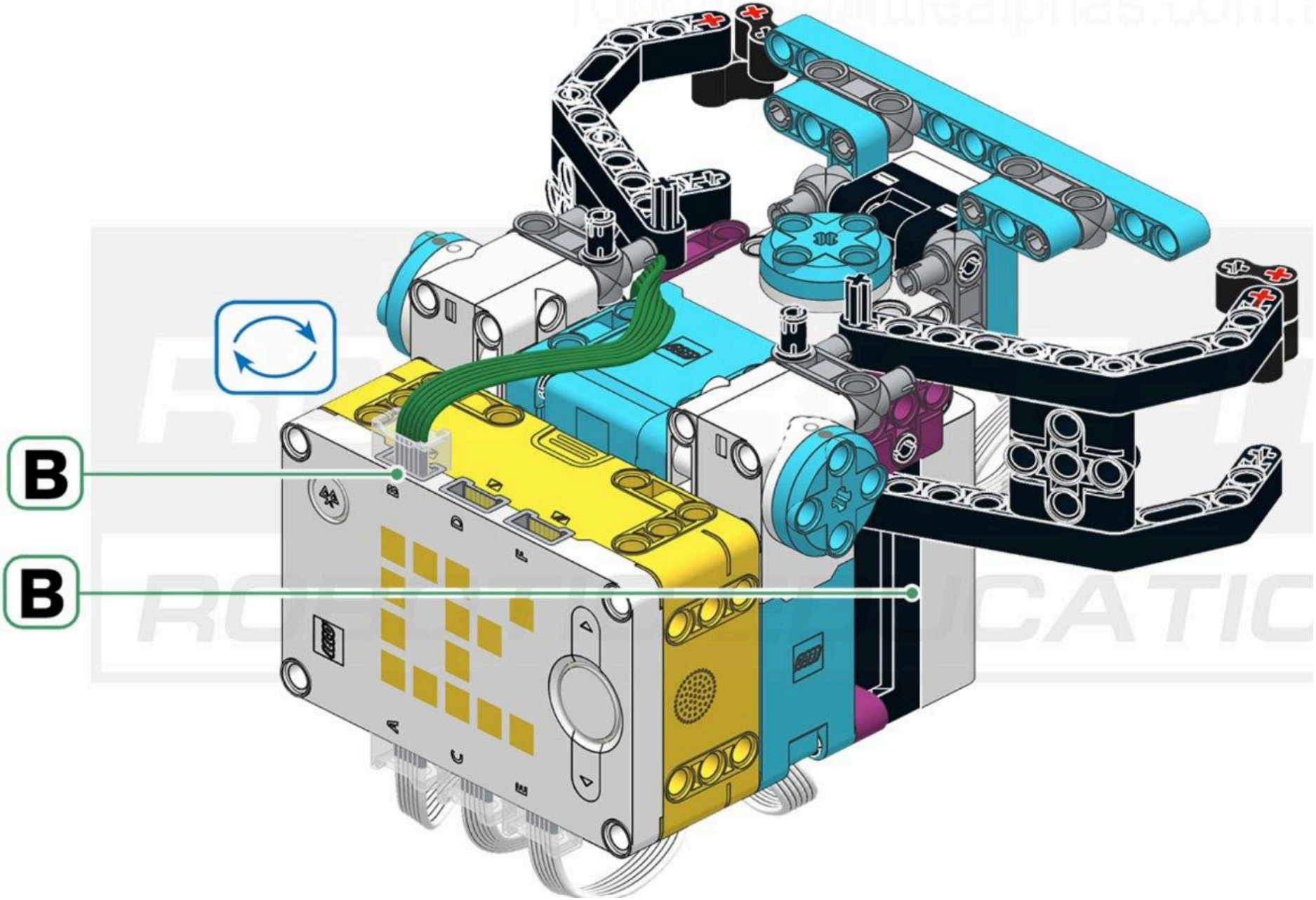


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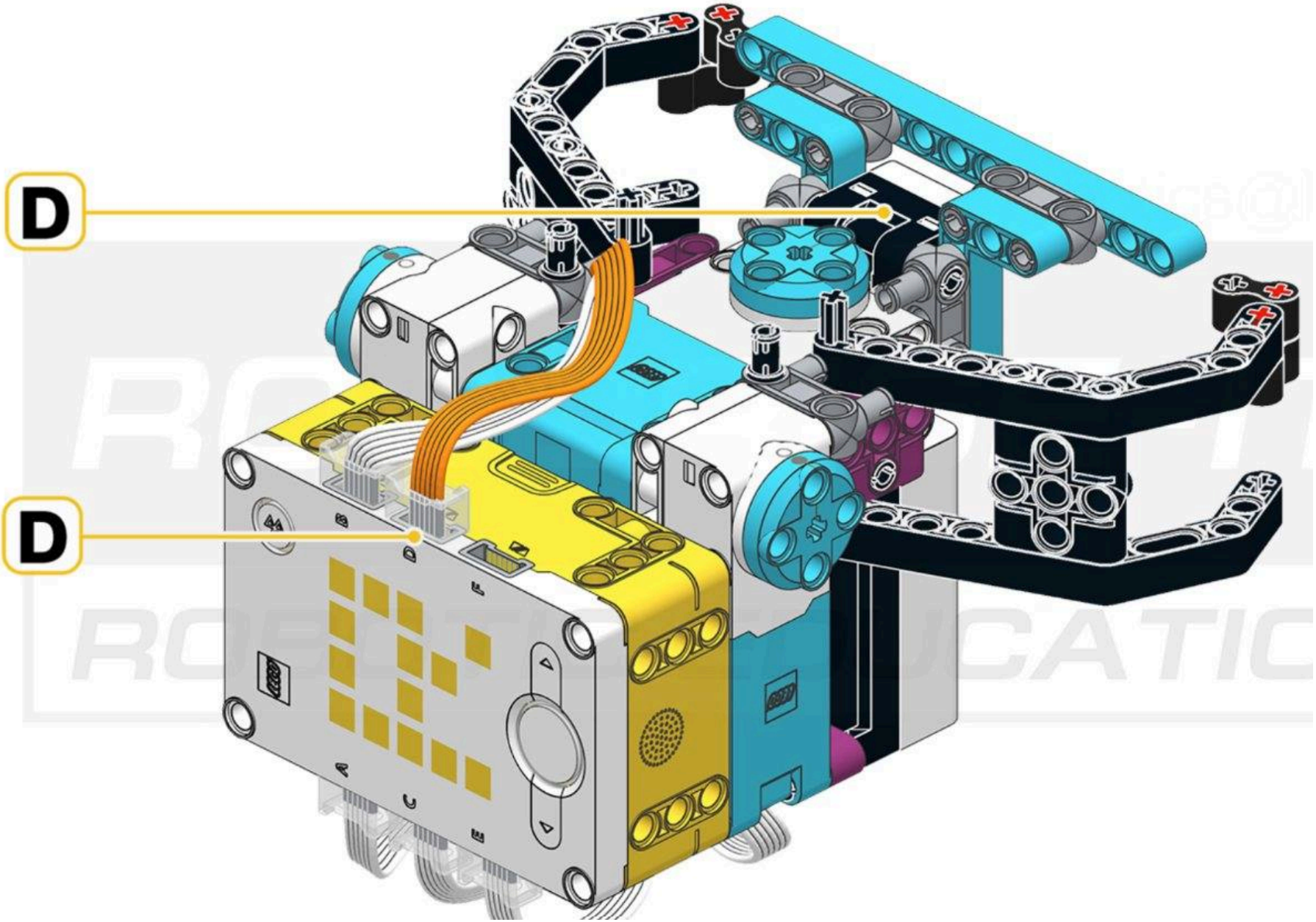


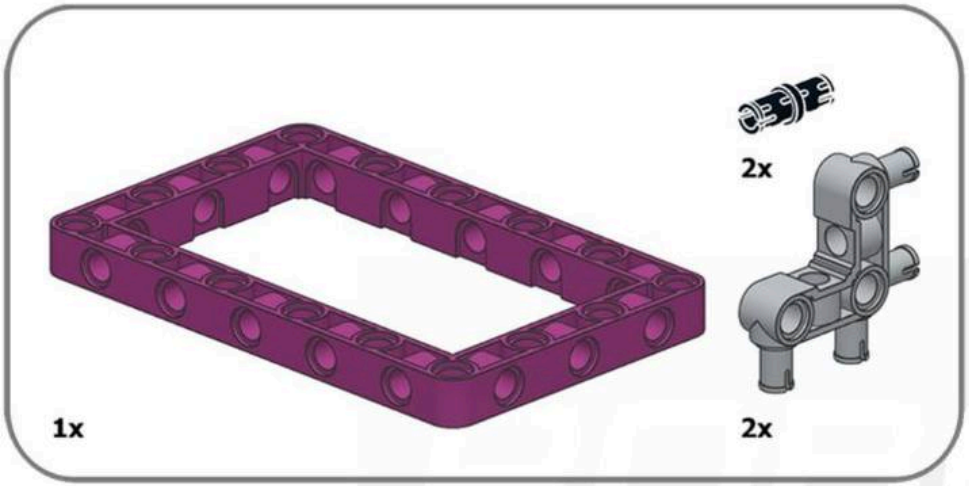
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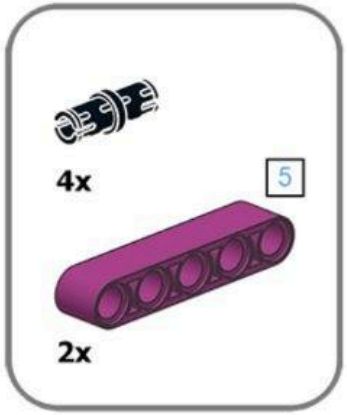




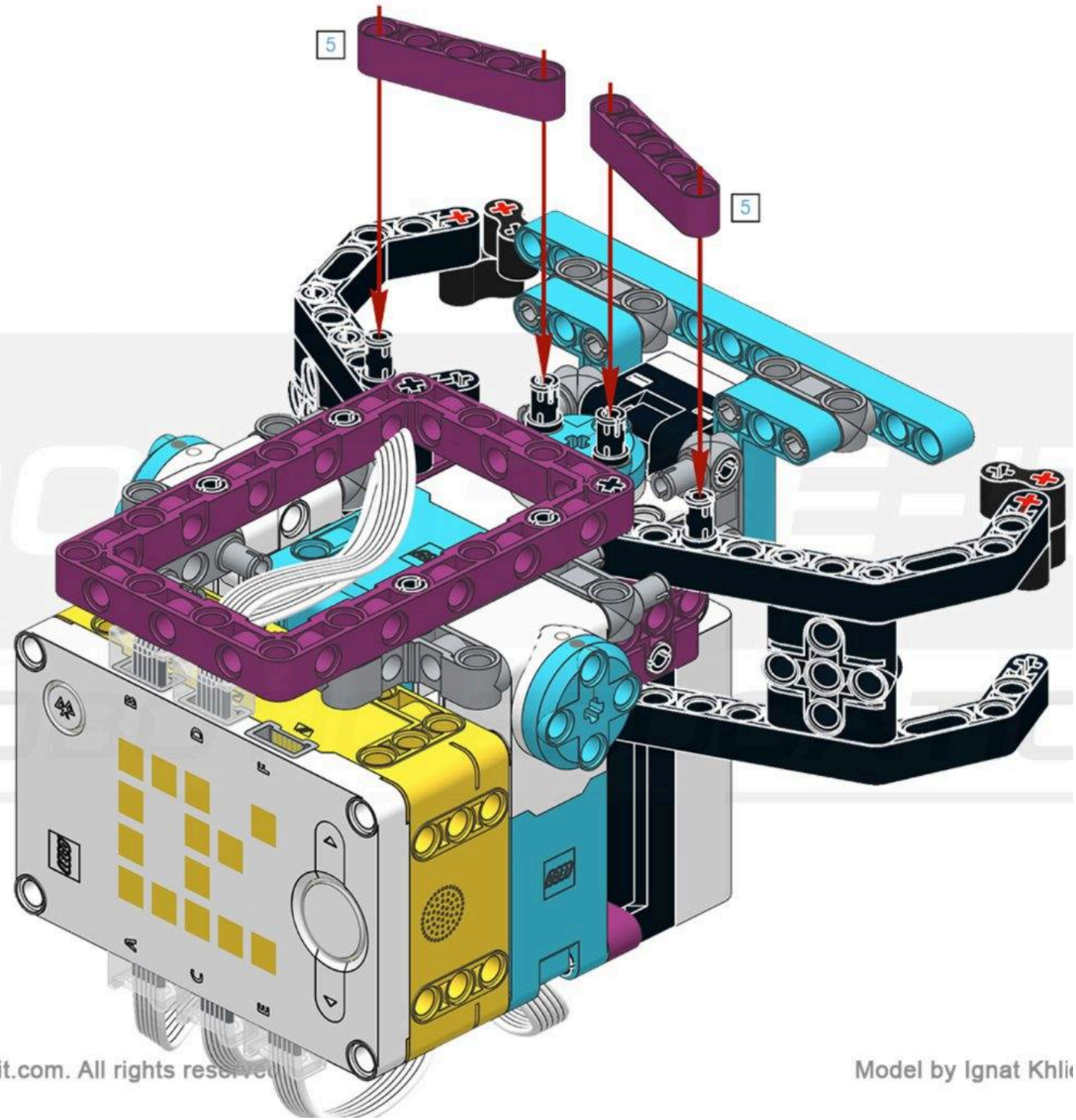
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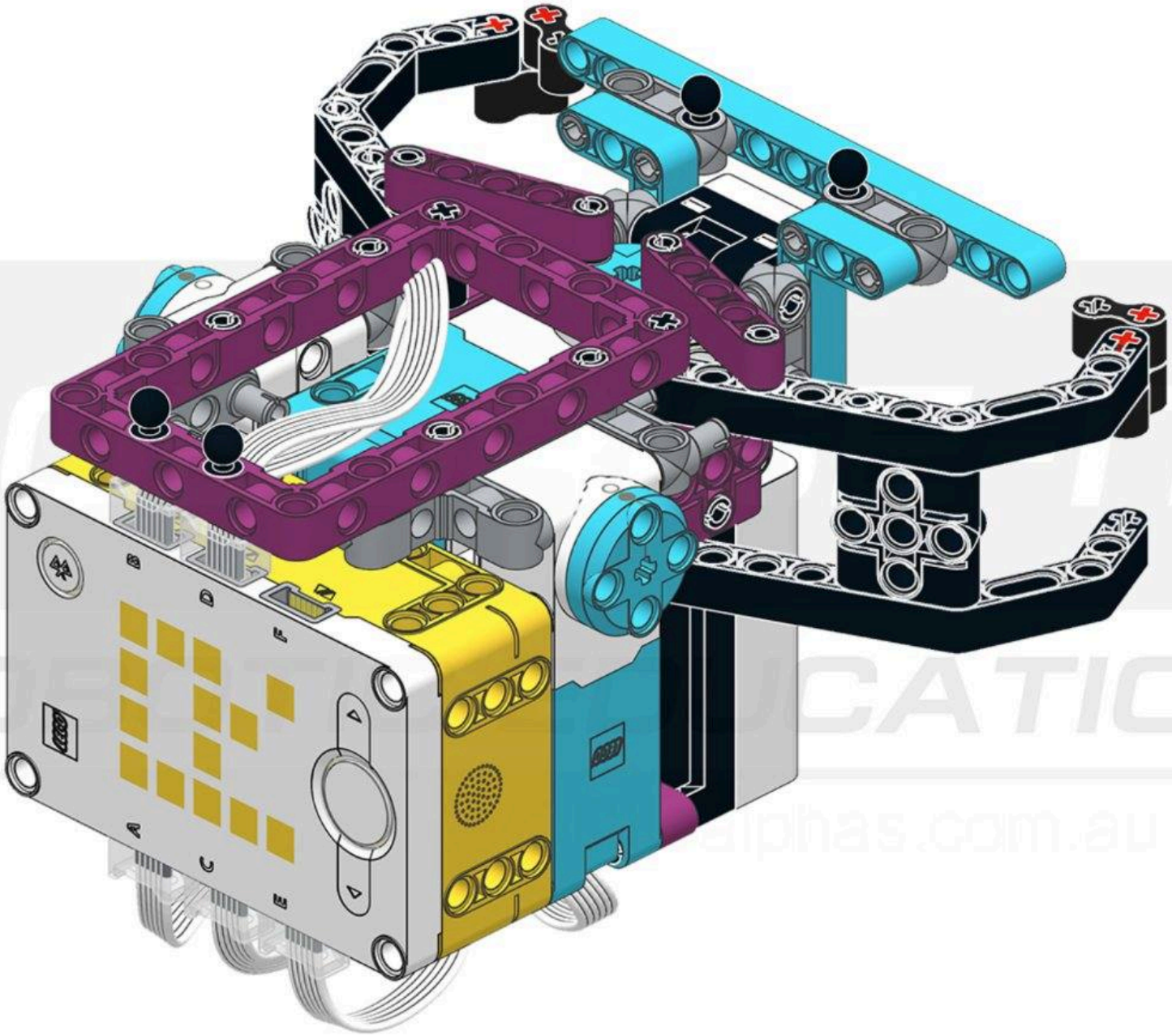


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Model by Ignat Khli

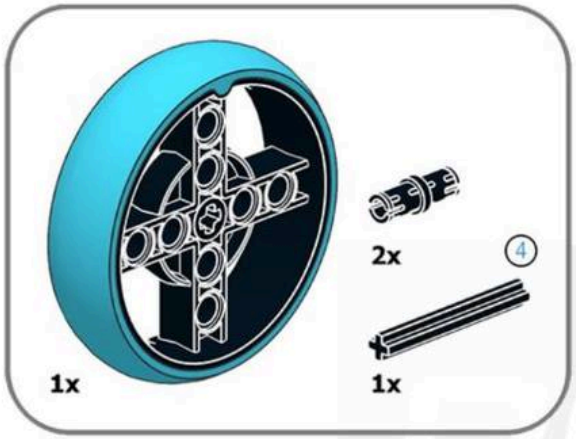


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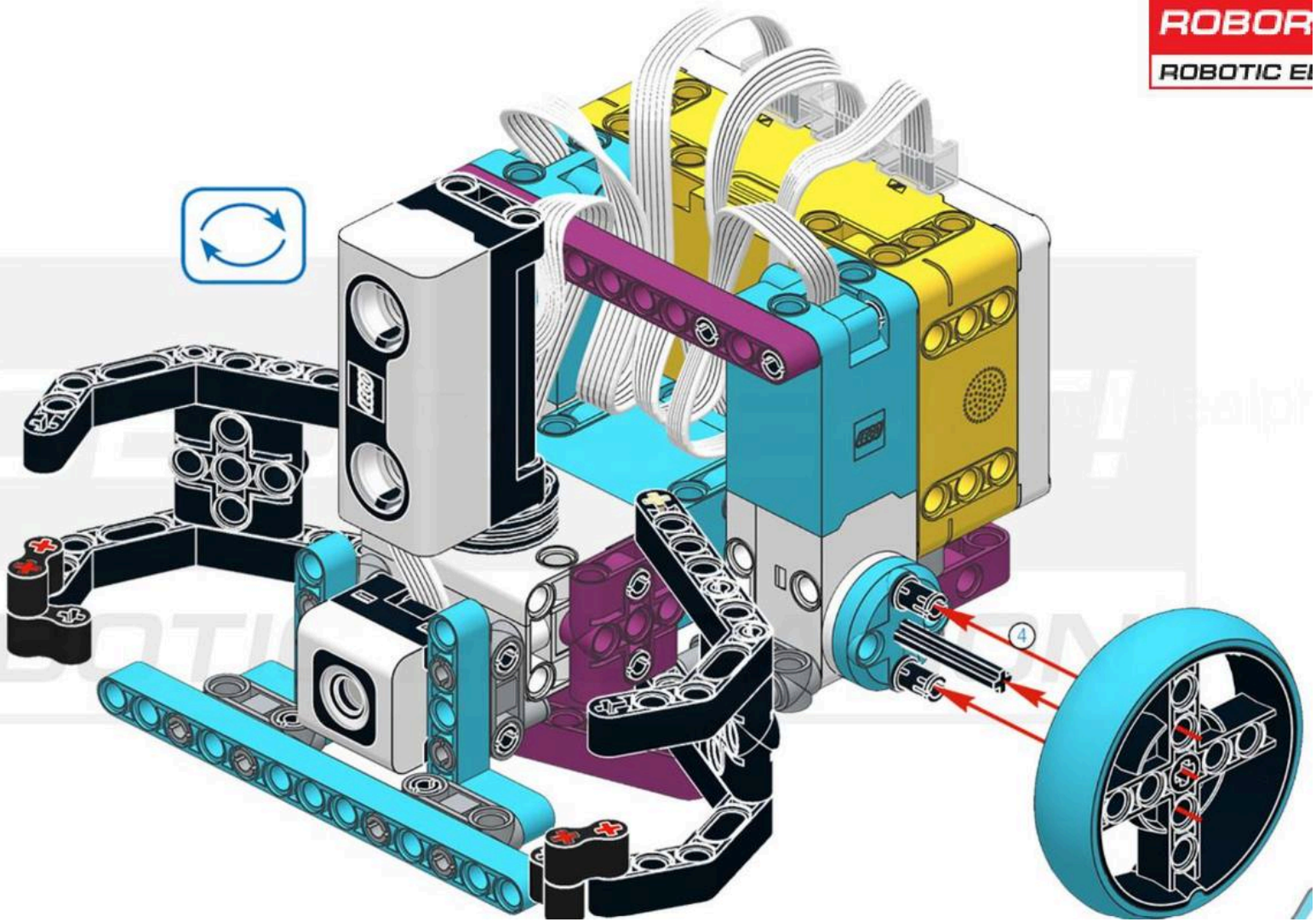


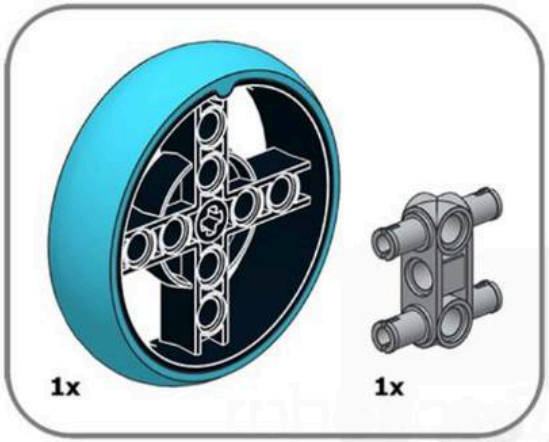
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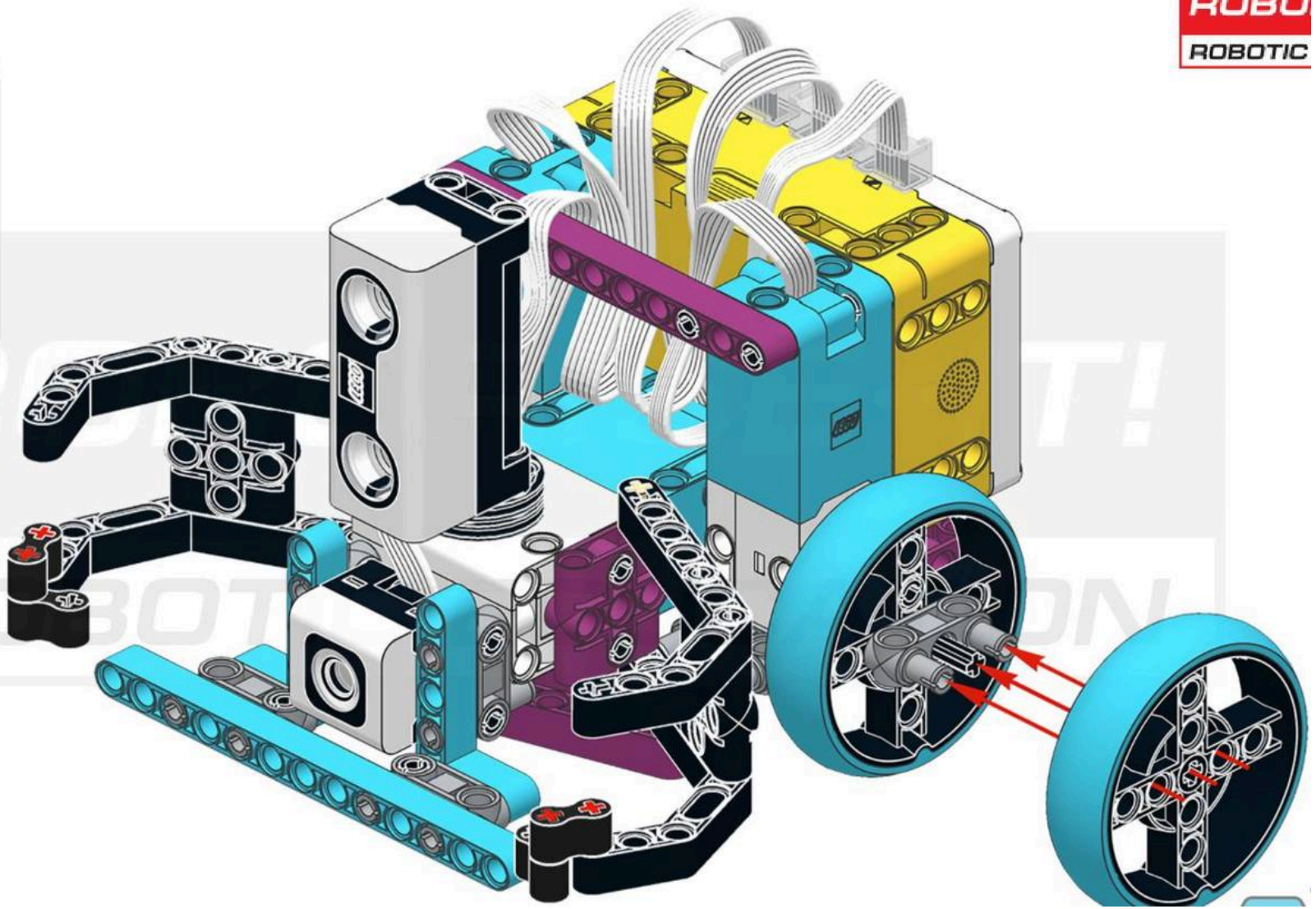


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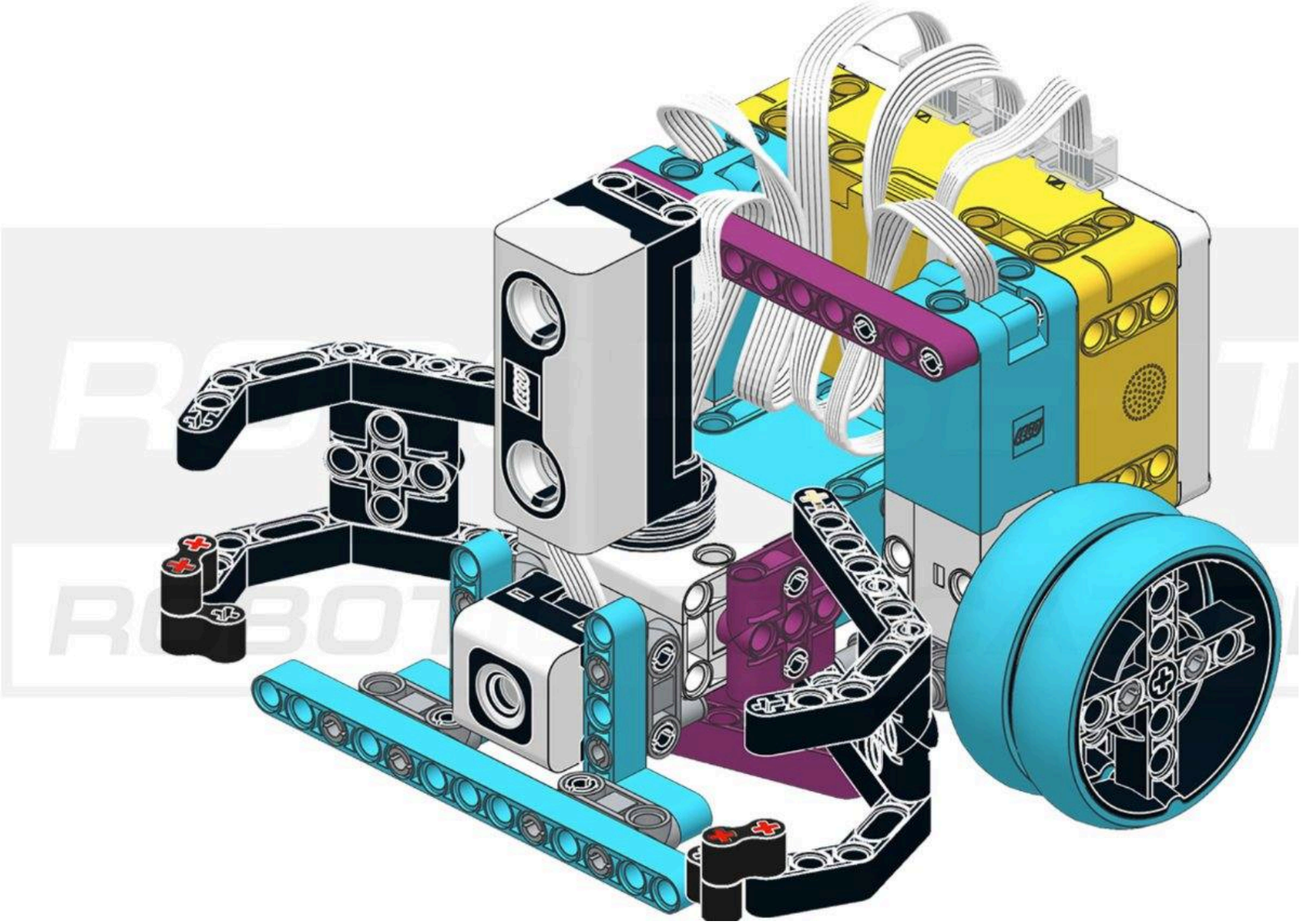


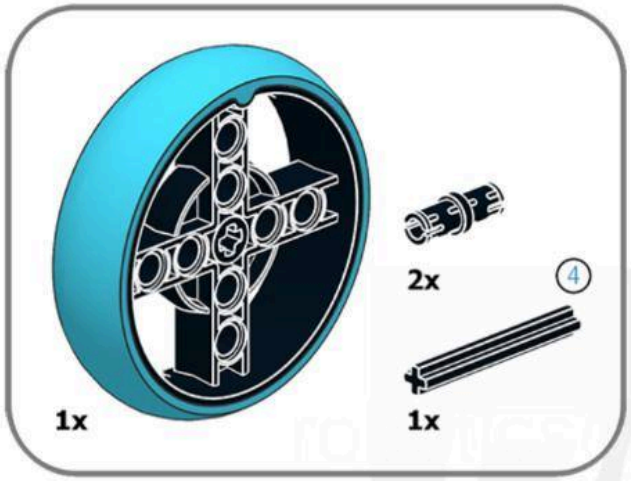


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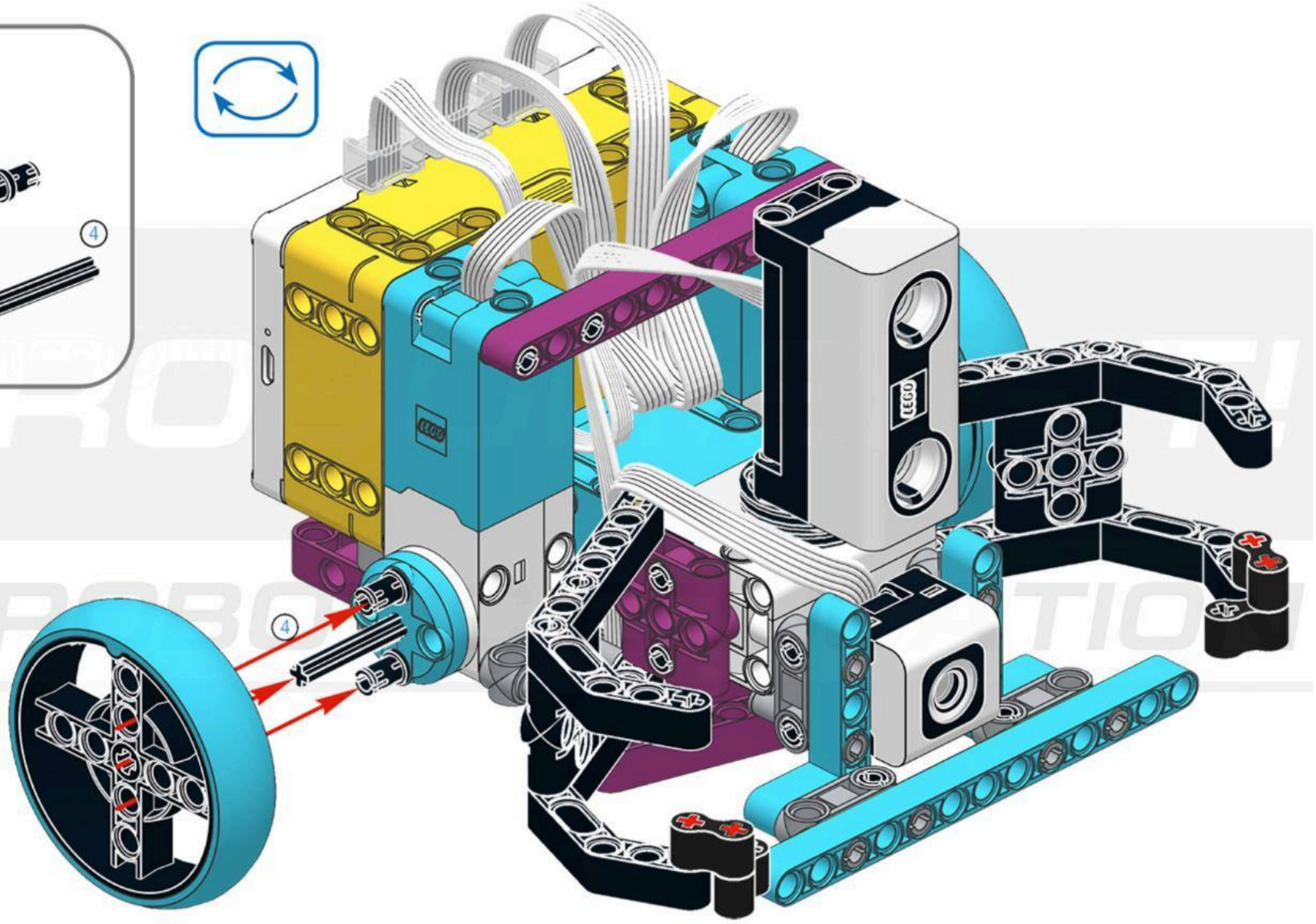


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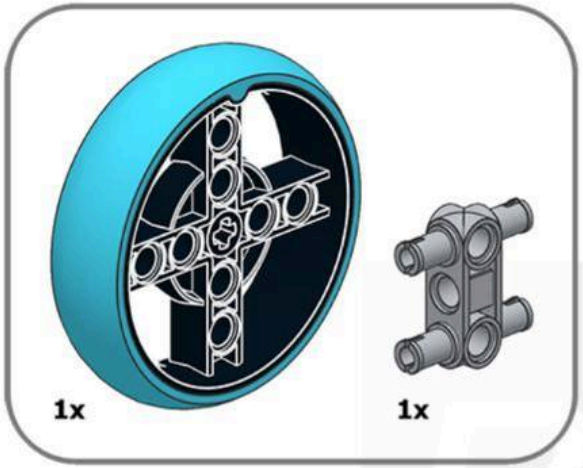




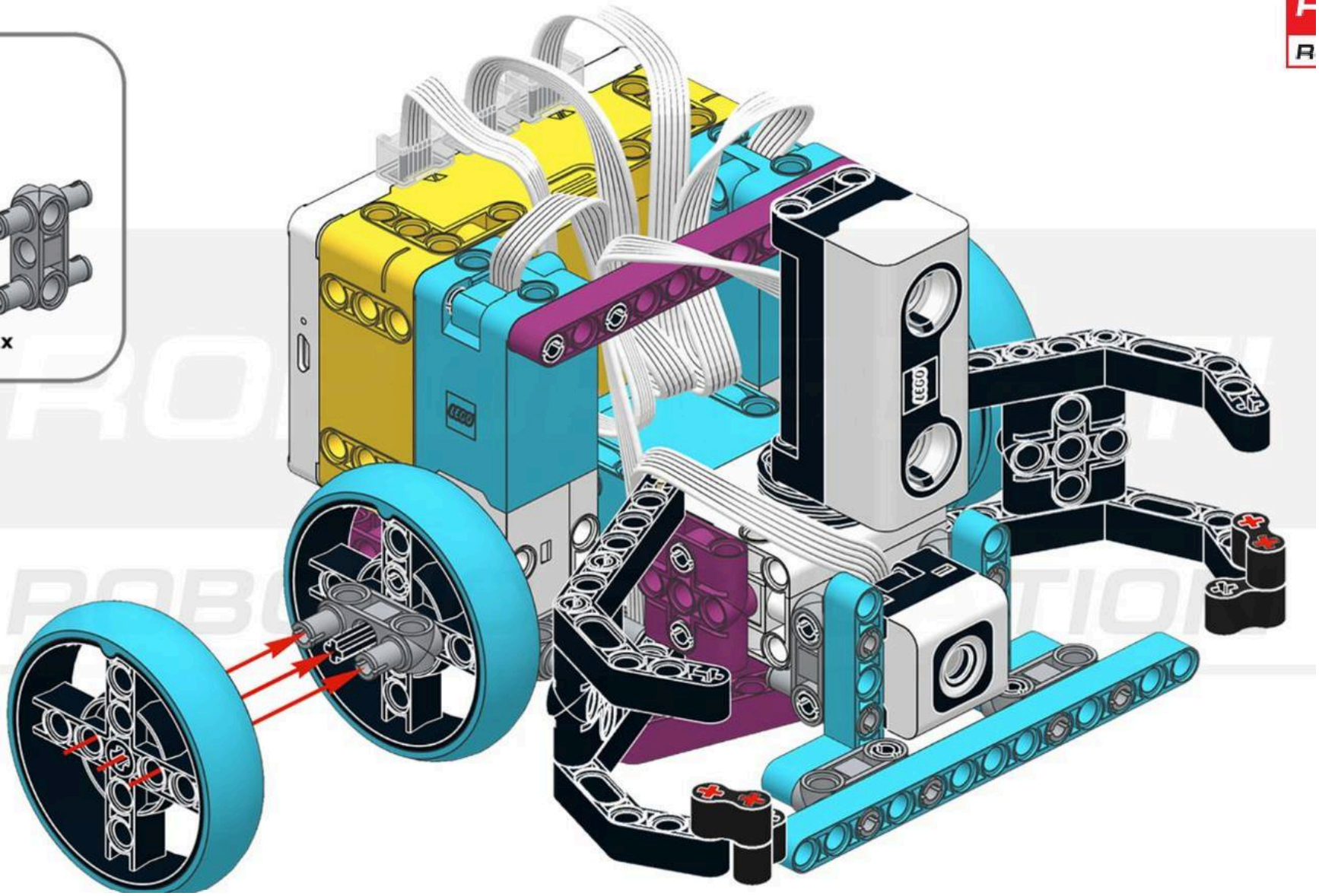
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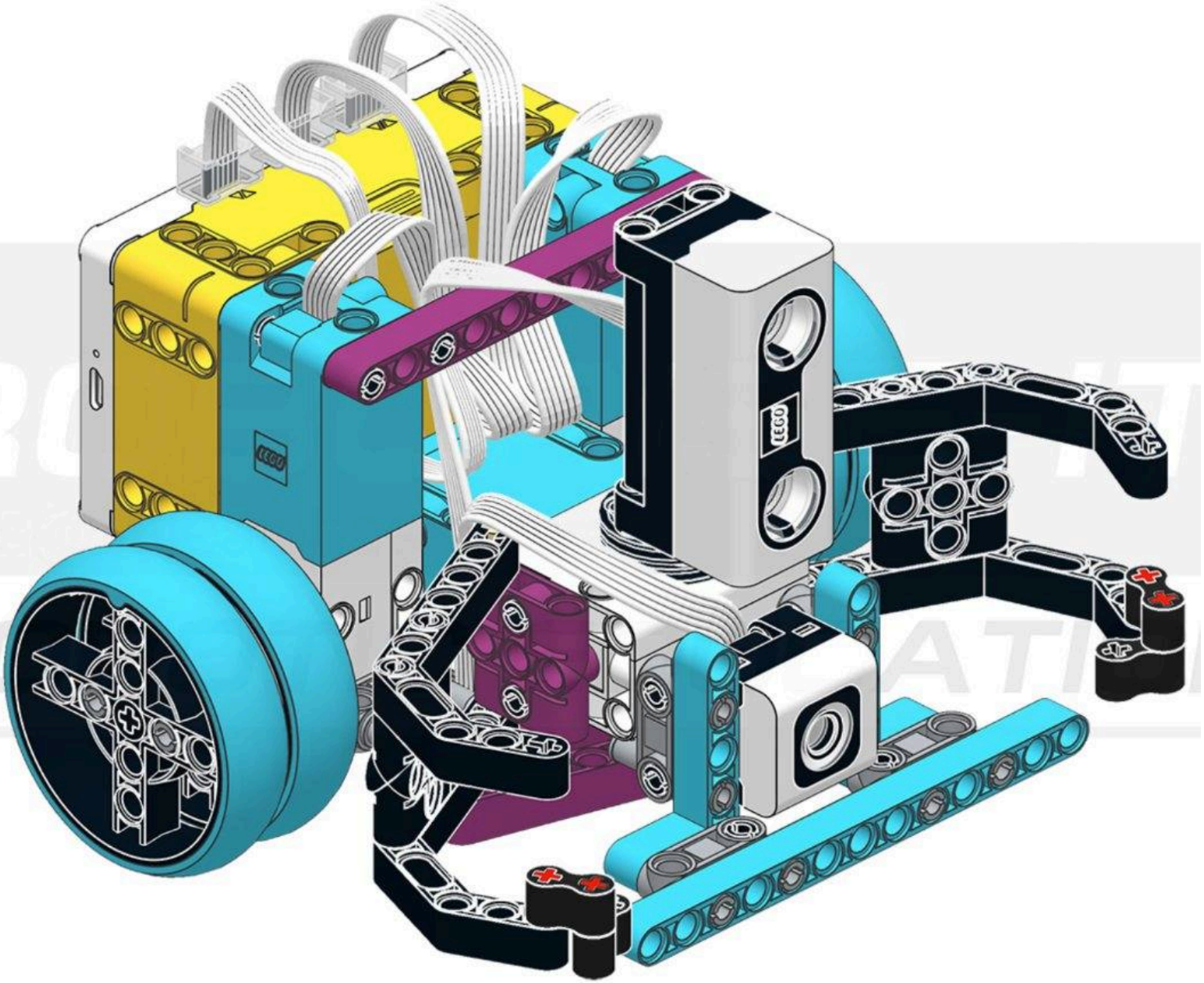
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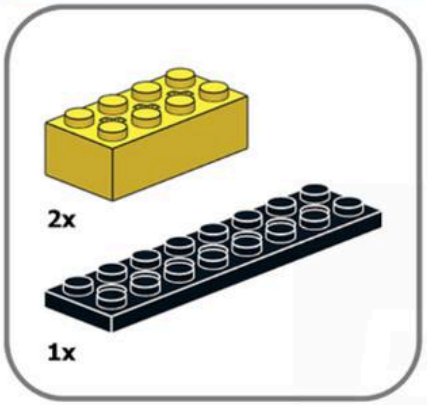


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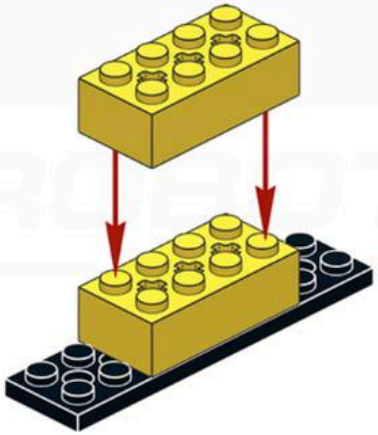


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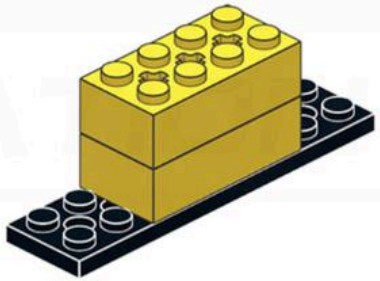




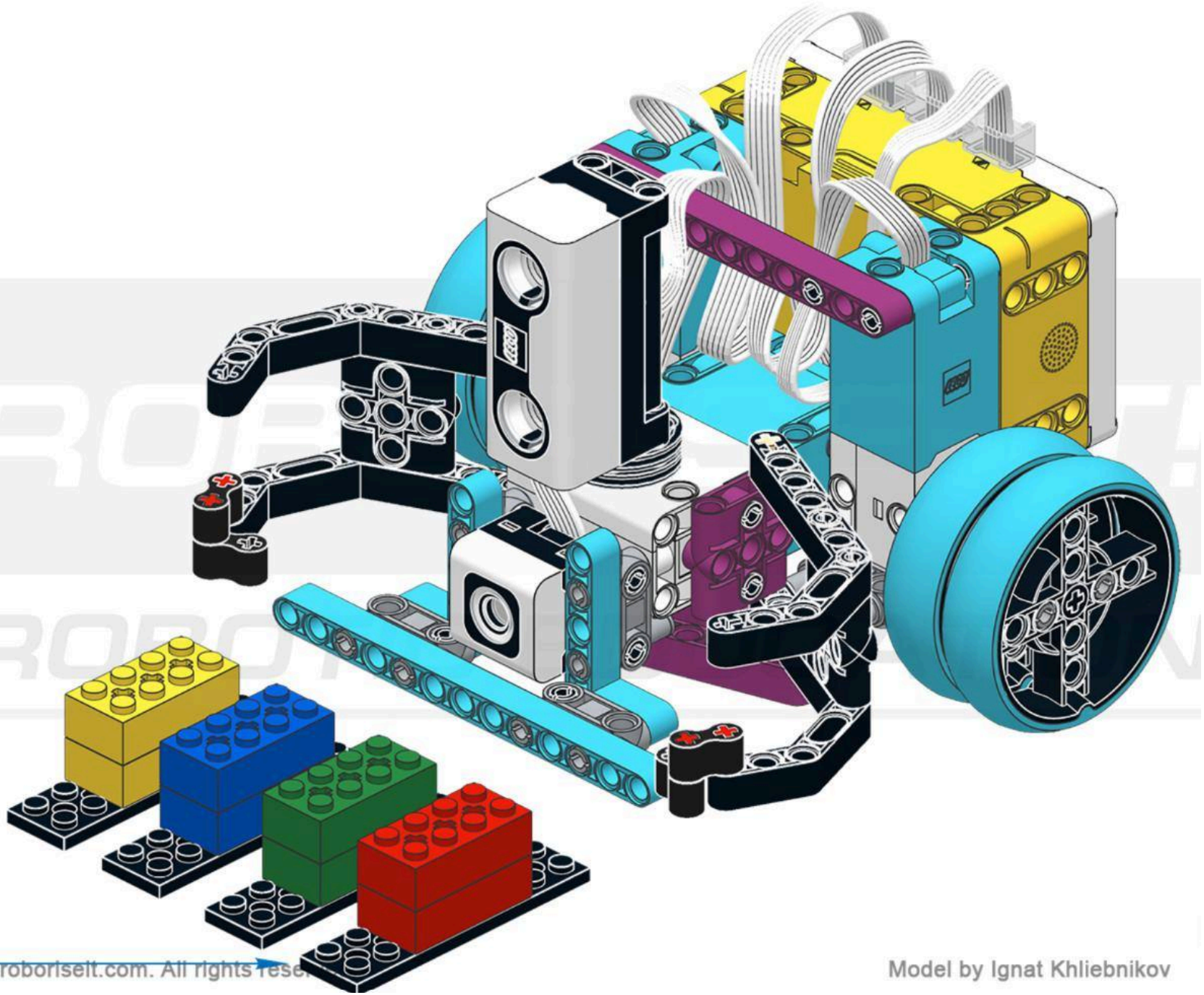
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2



47



47/47

0

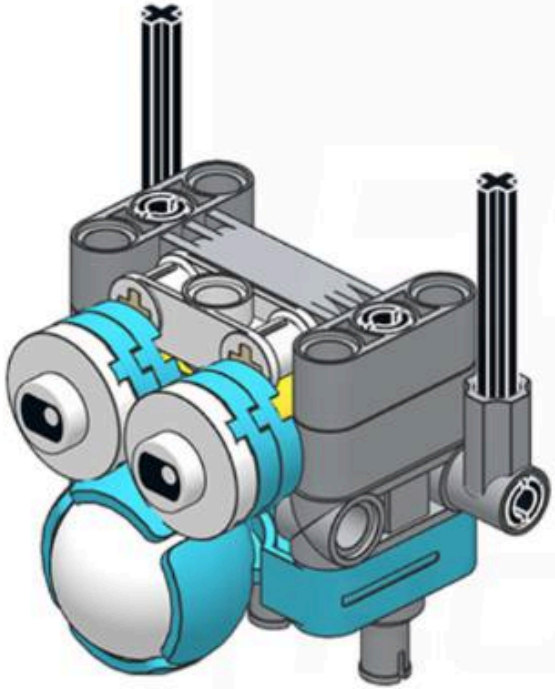
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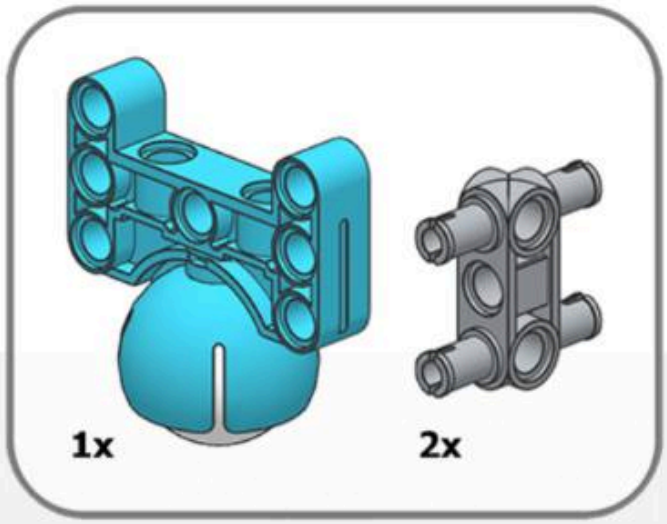
Model by Ignat Khliebnikov

Advanced version

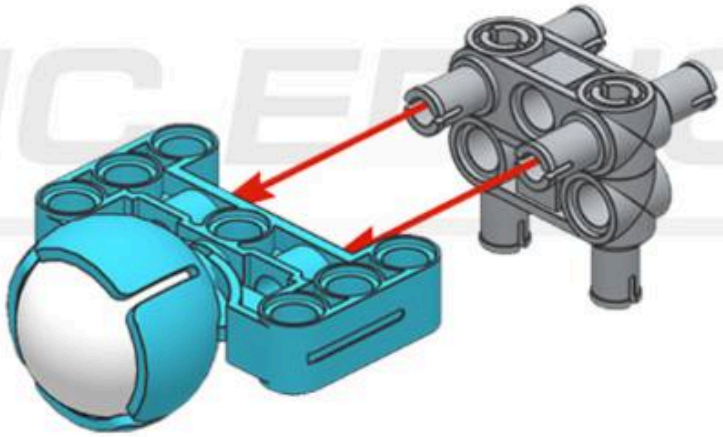
Build a robot 'head' with antennas and visual surveillance sensors

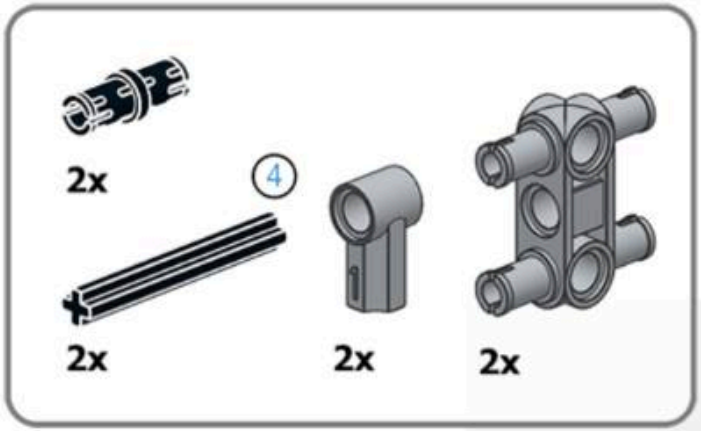




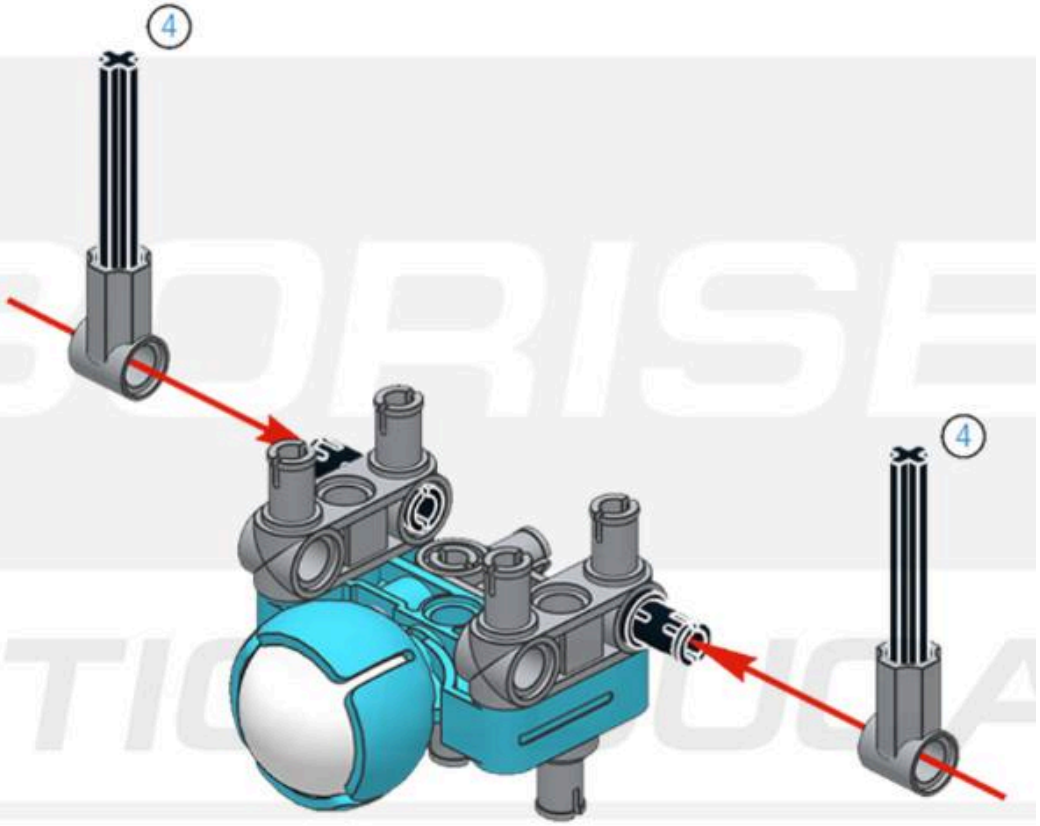


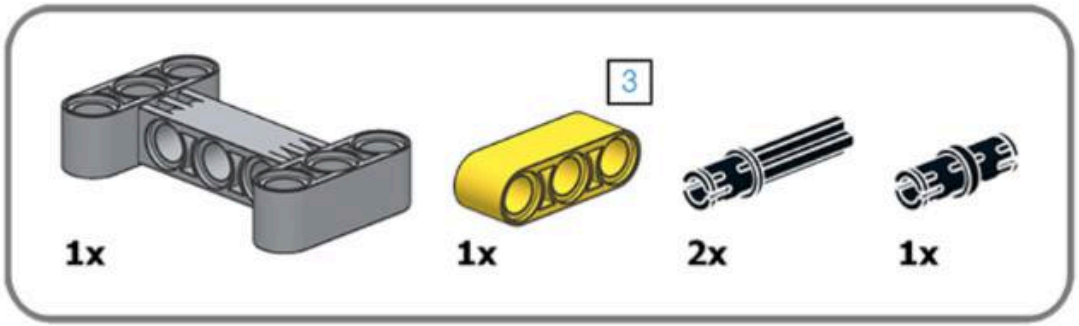
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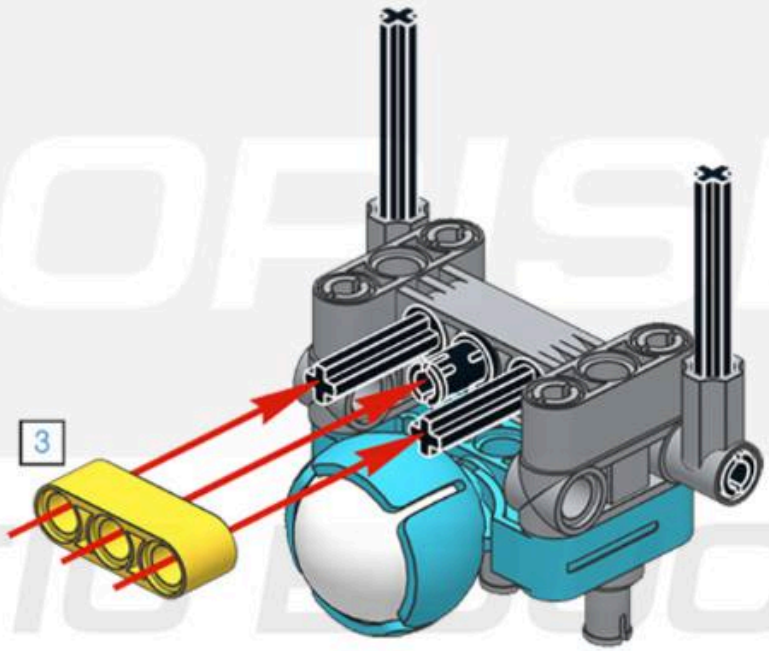


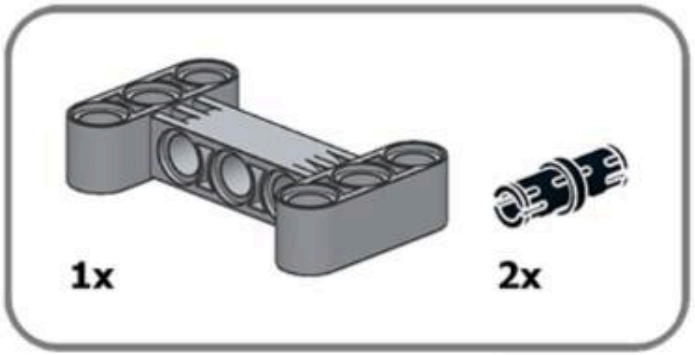
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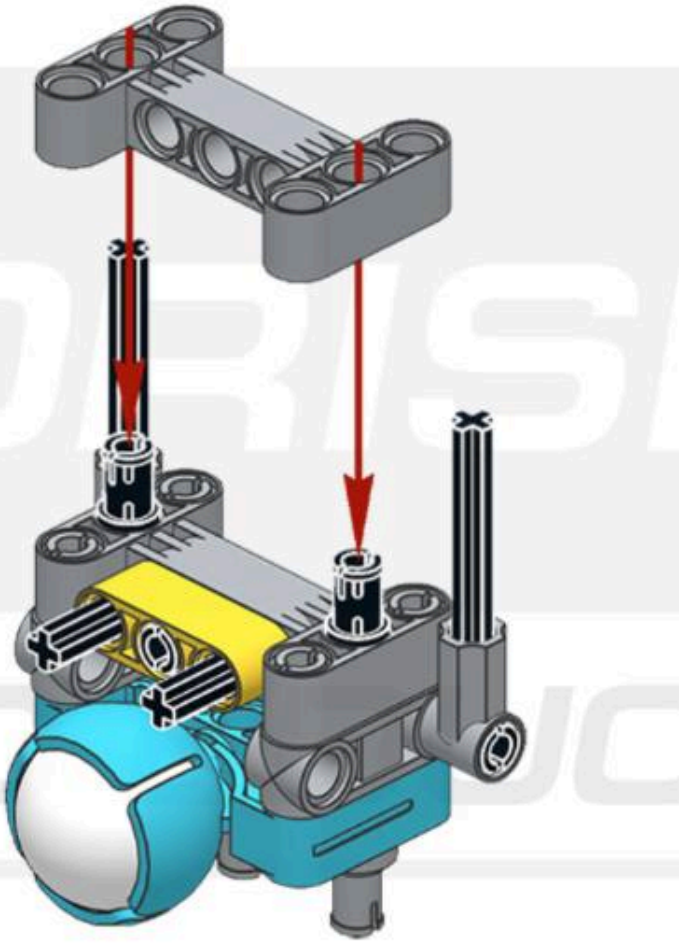


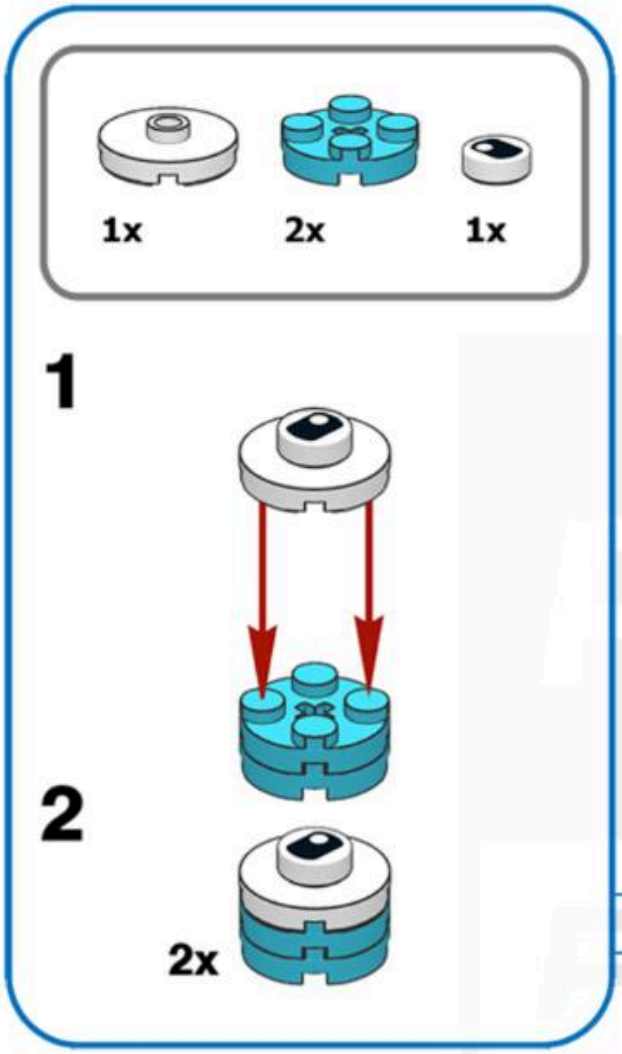
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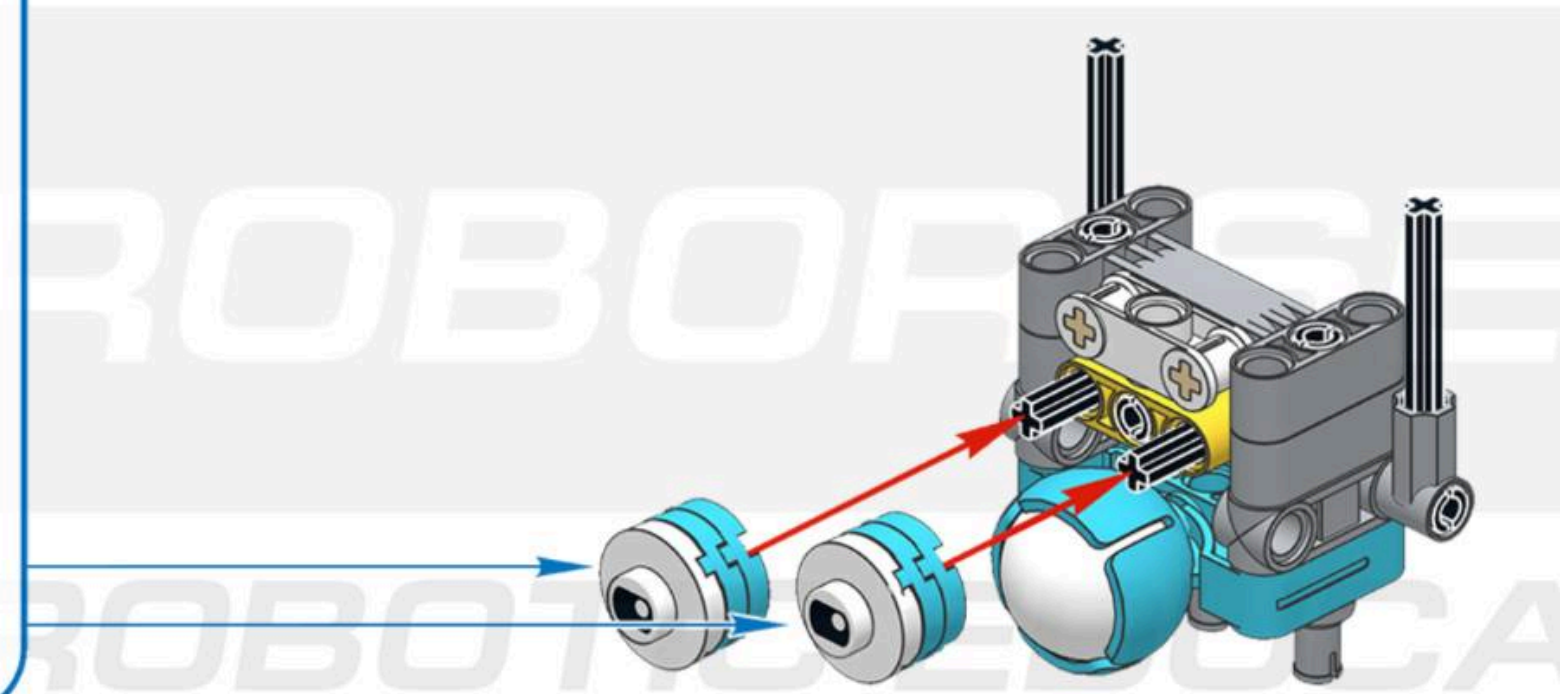




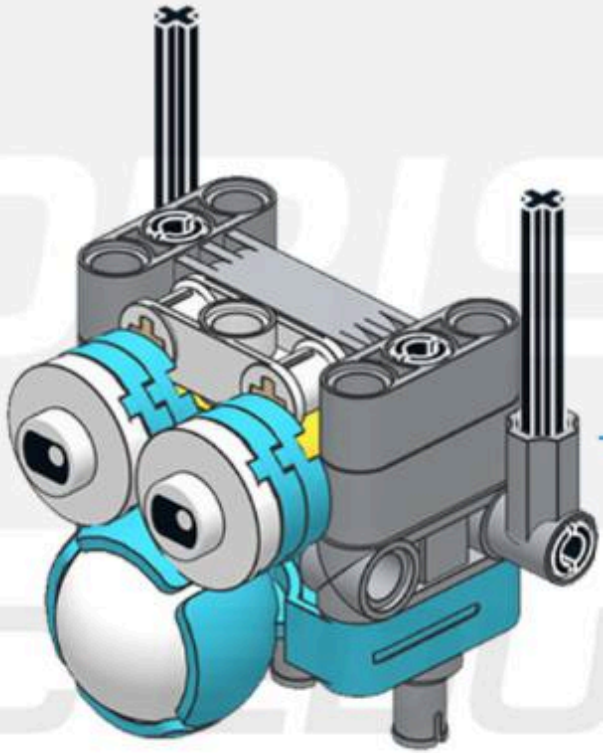
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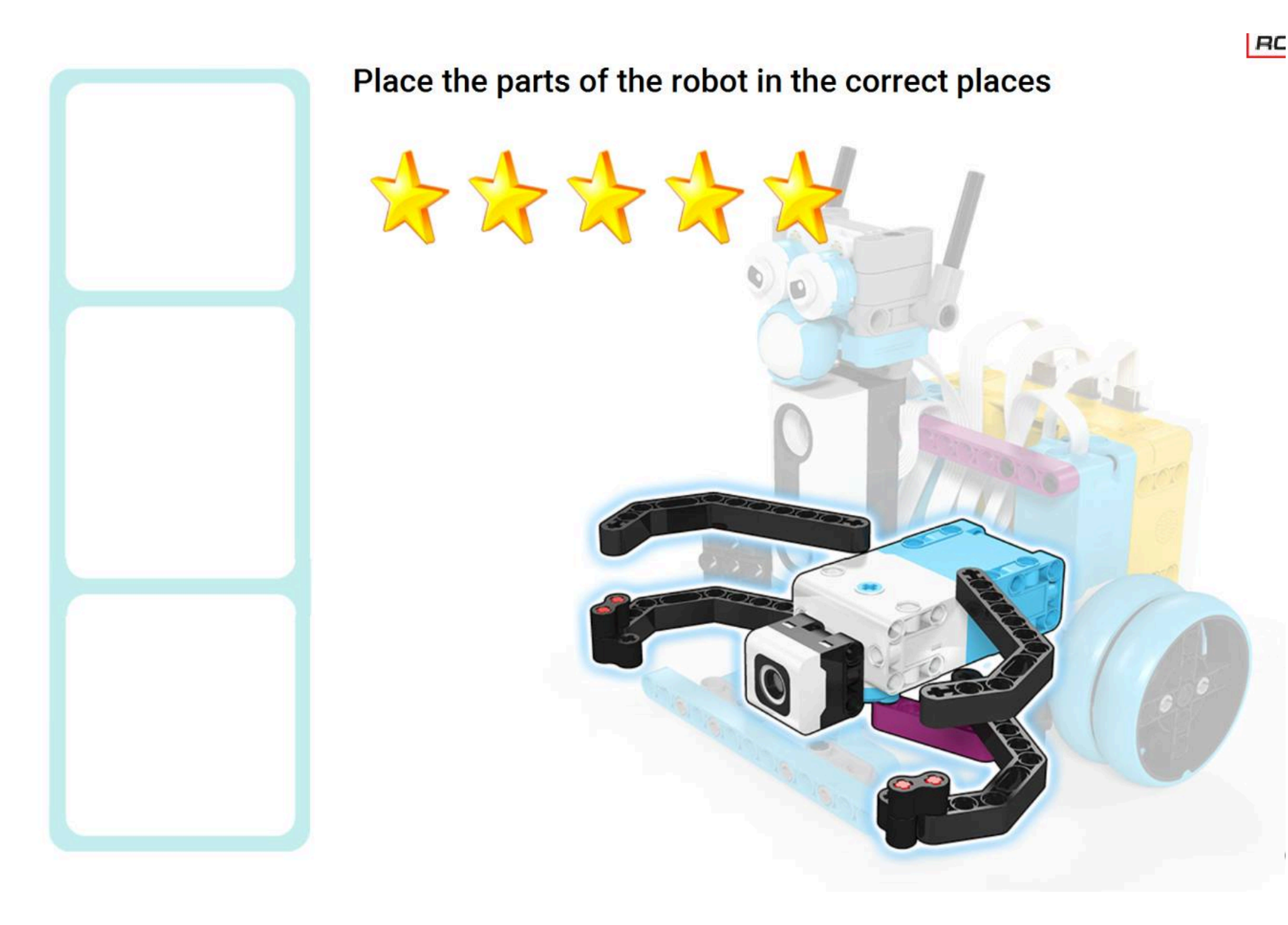

1x
2x
1x
1
2
2x



55

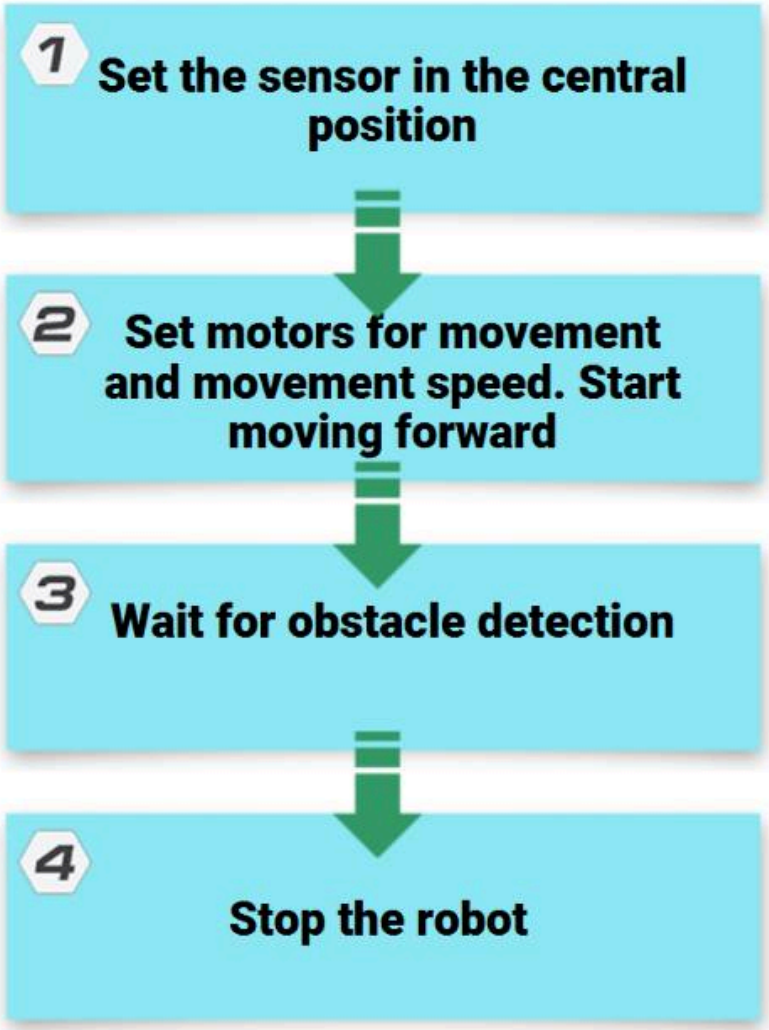


Place the parts of the robot in the correct places

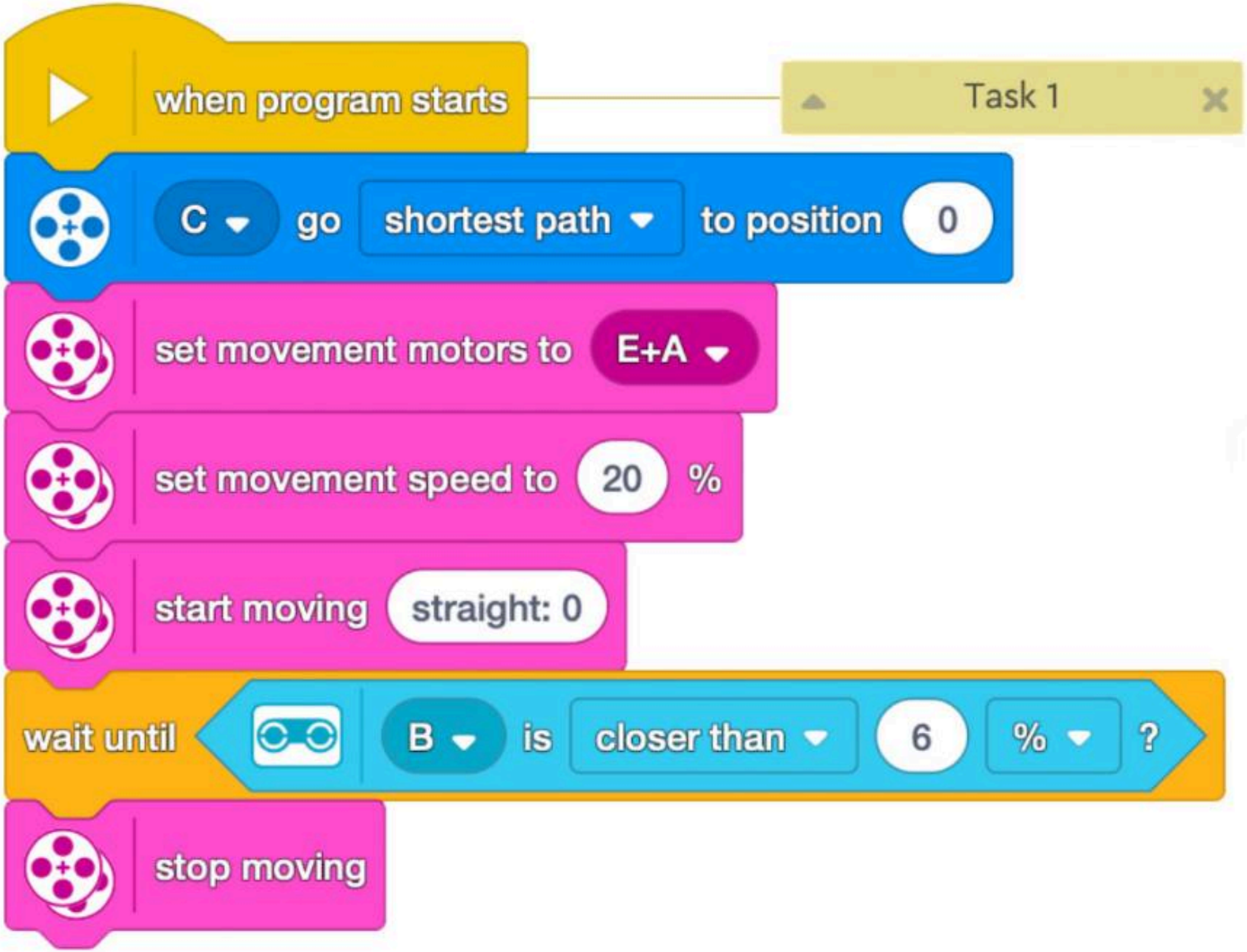


Program the robot to move forward and stop before an obstacle.

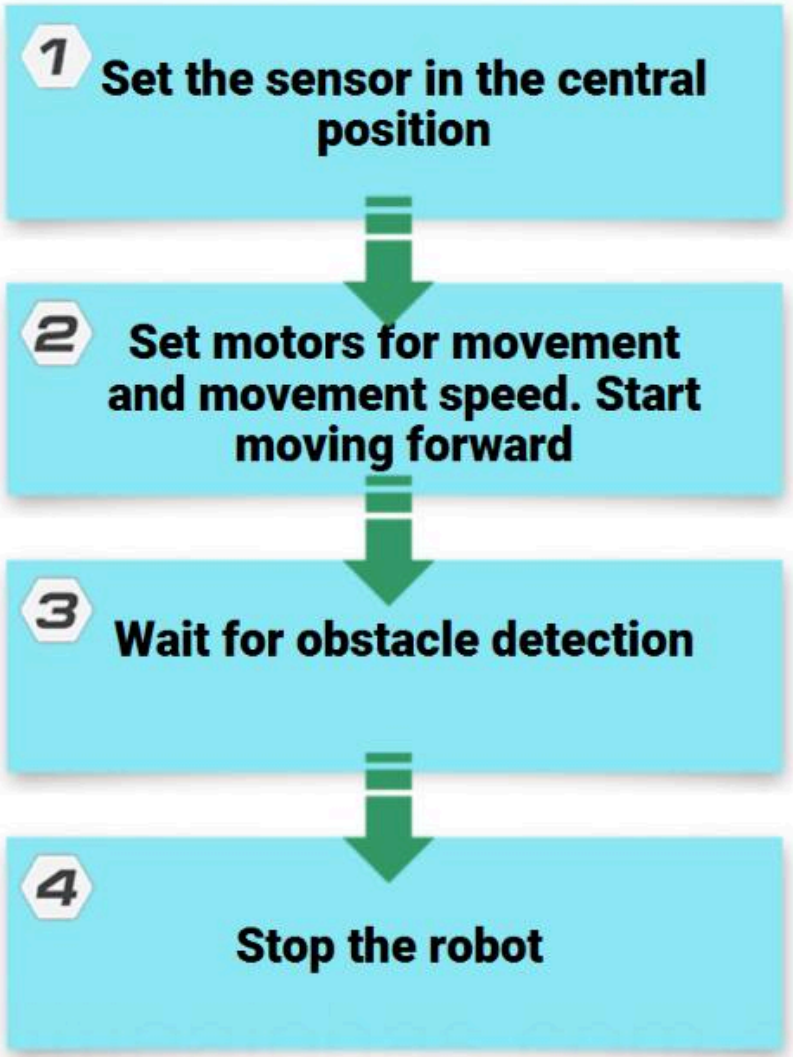
To perform the task, the robot must work according to the following algorithm:



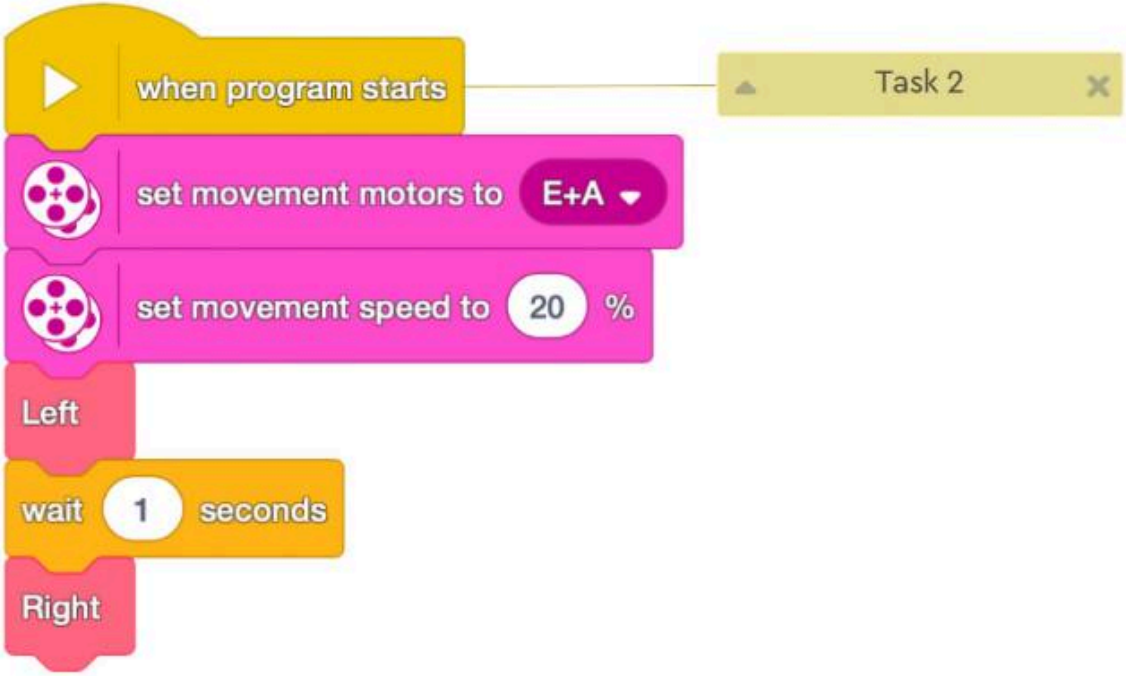
The simplest implementation of such an algorithm looks like this:



To perform the task, the robot must work according to the following algorithm:



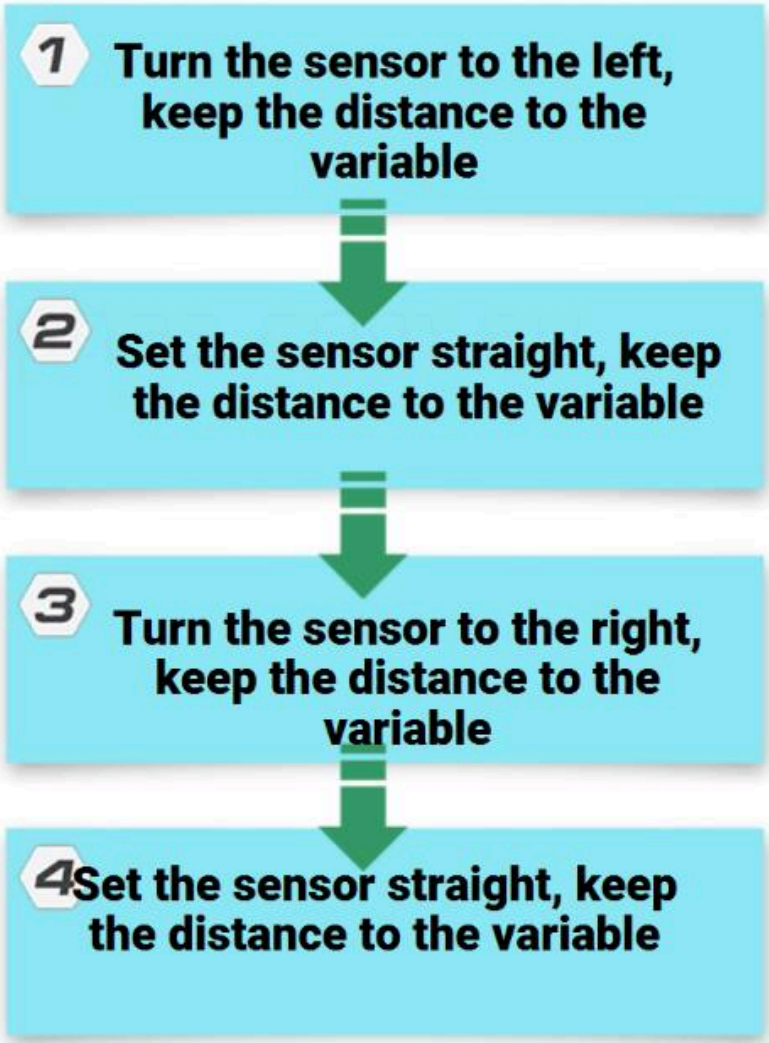
An improved program might look like this:







To perform the assigned task, the program must work according to the following algorithm:

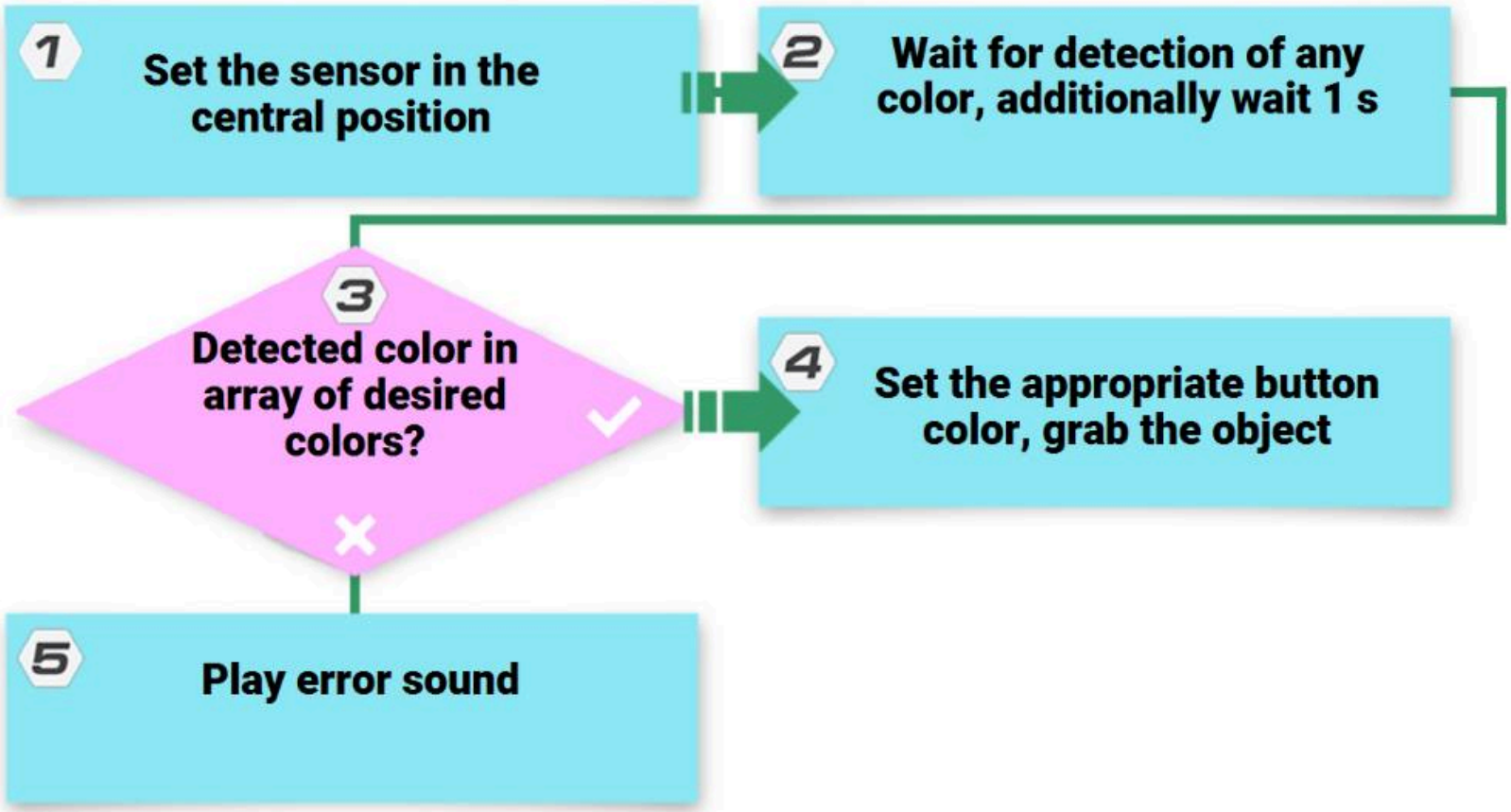


Use messaging and add the following part of the program:

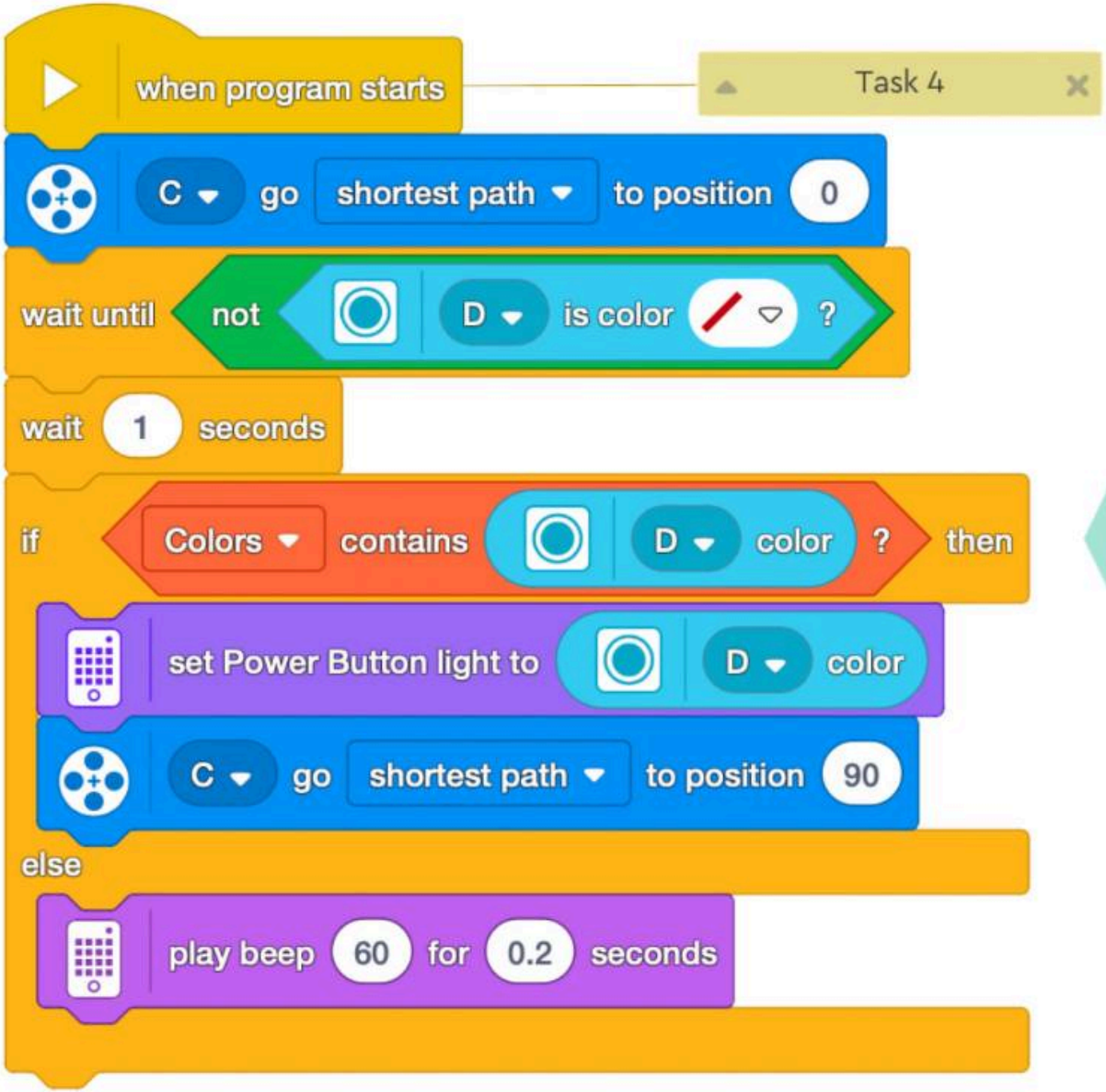




The algorithm of the program should contain processing of two variants of events: when the detected color is in the array of required colors, and when the color is not in the array



Write the following program:



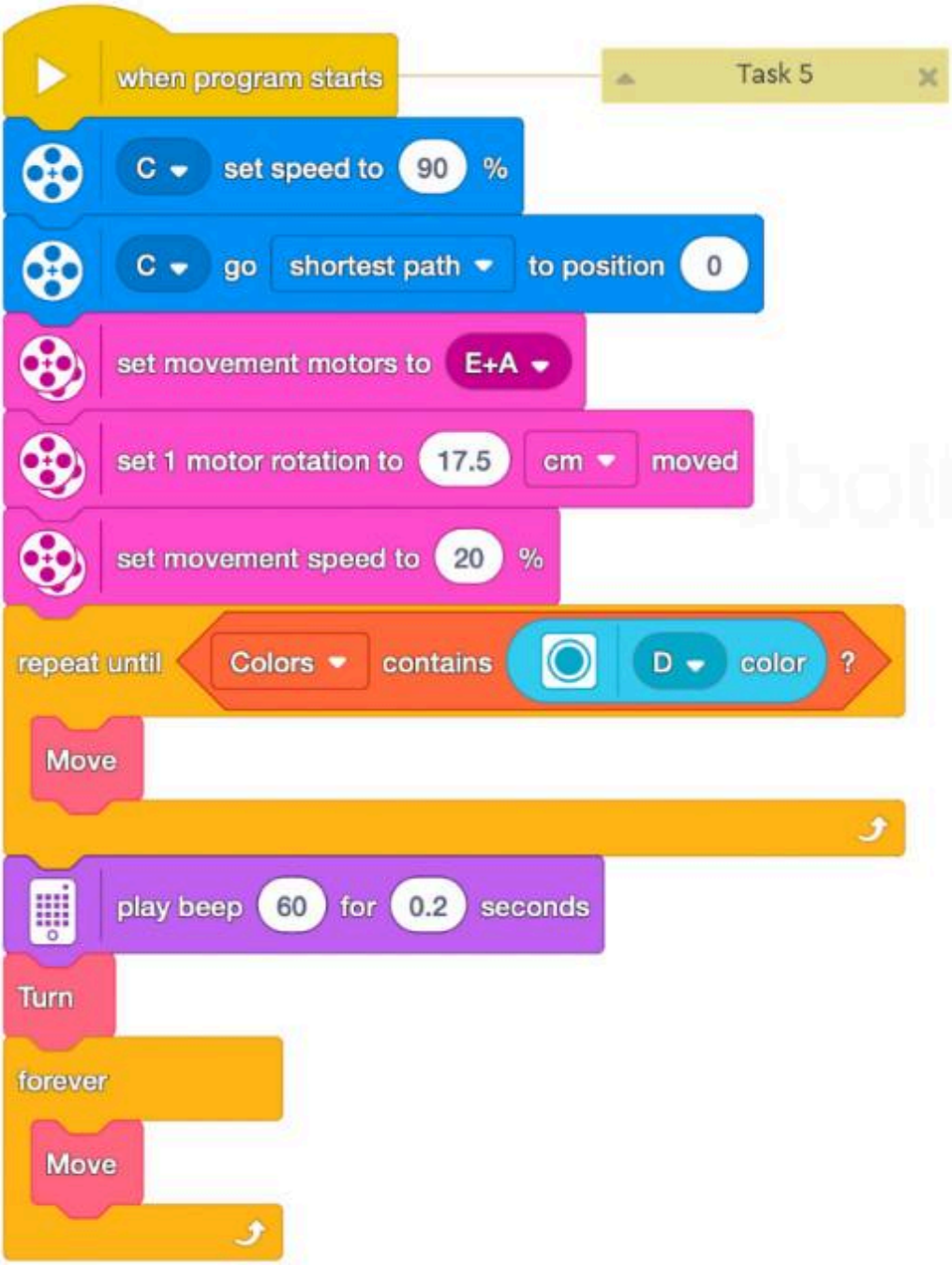
The robot captures only those objects whose color number is in the 'Colors' array

Create an array and add to it the numbers of the colors that the robot should capture:

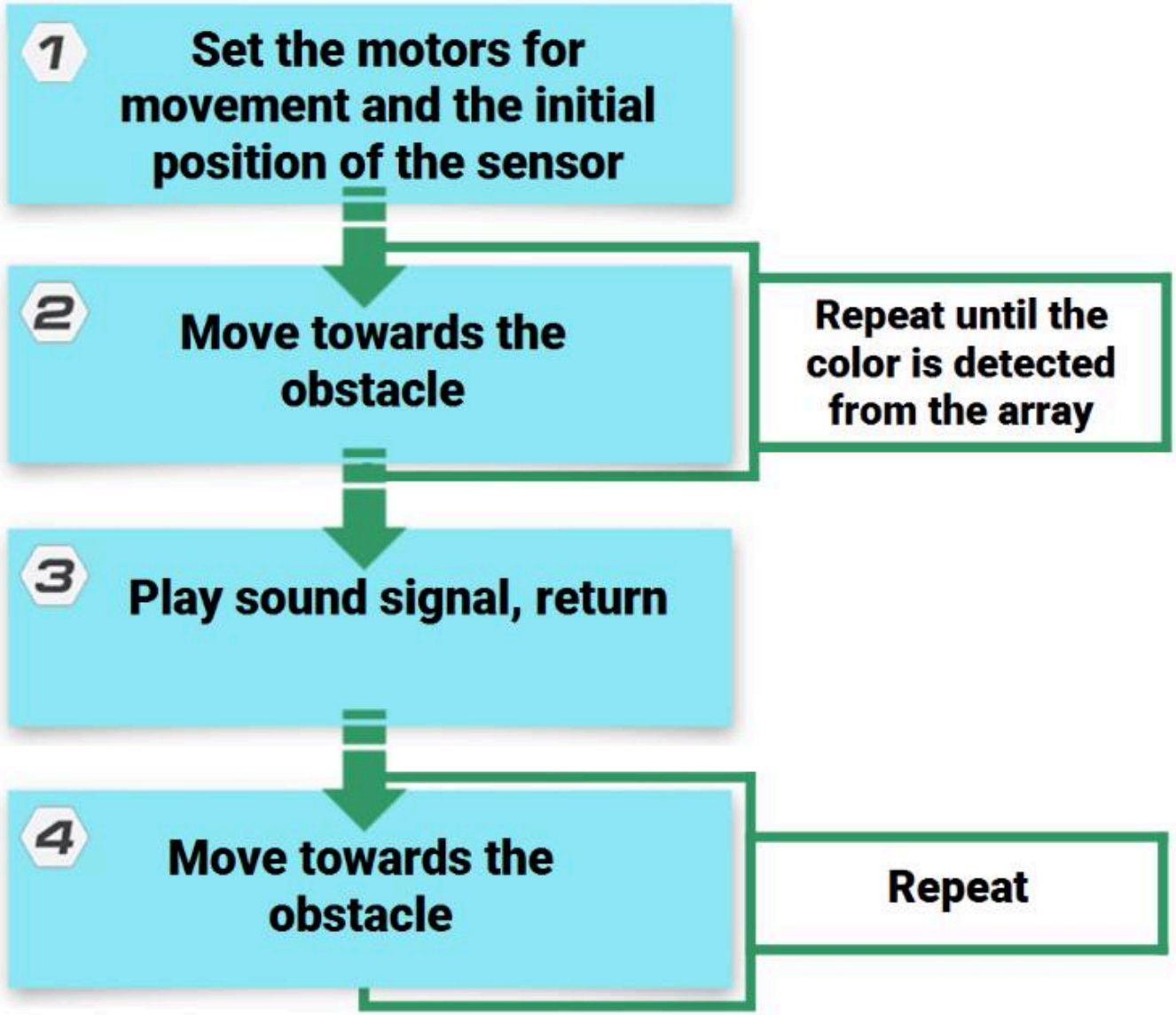


robotic

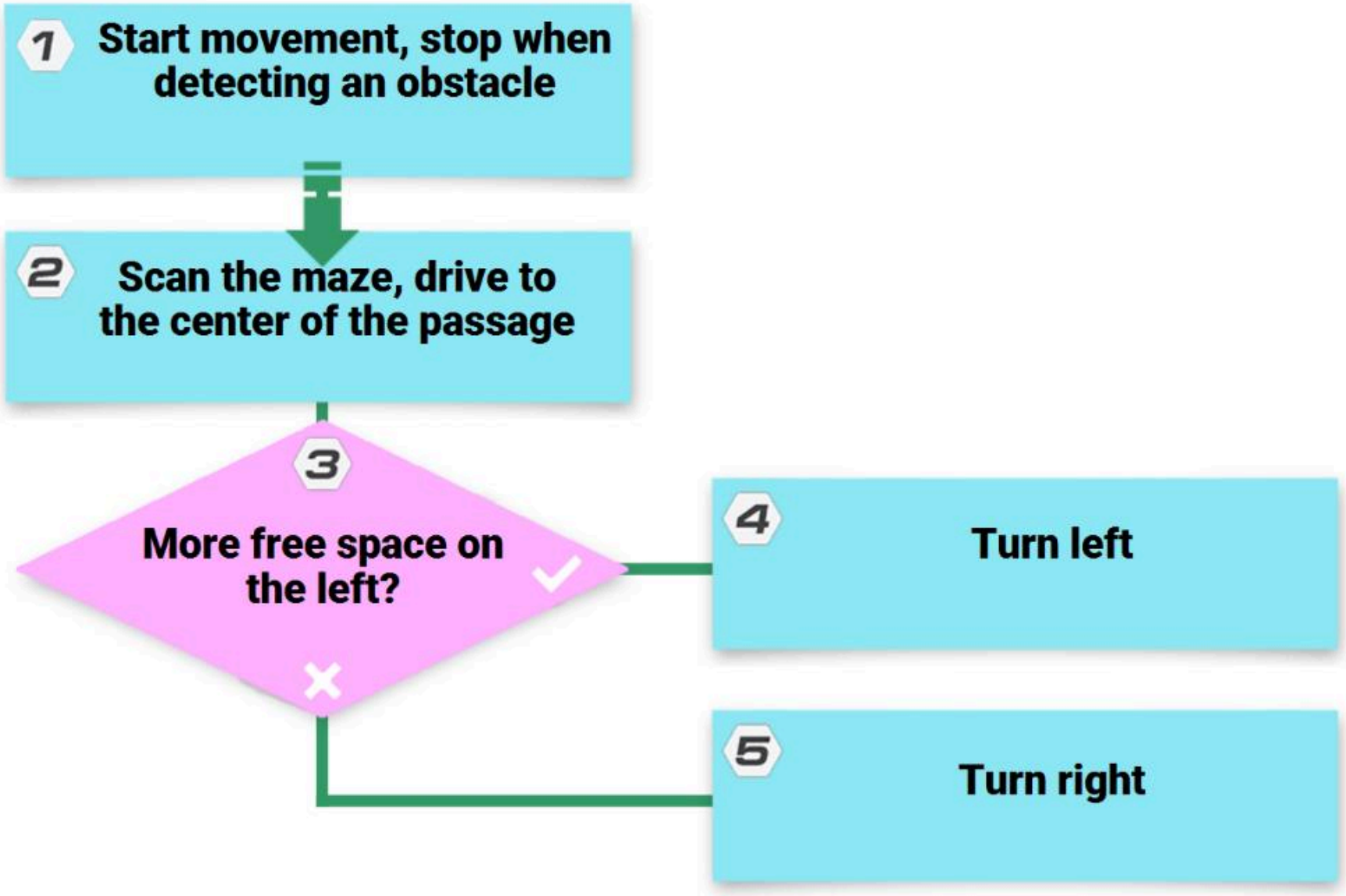
The main program execution cycle is as follows:



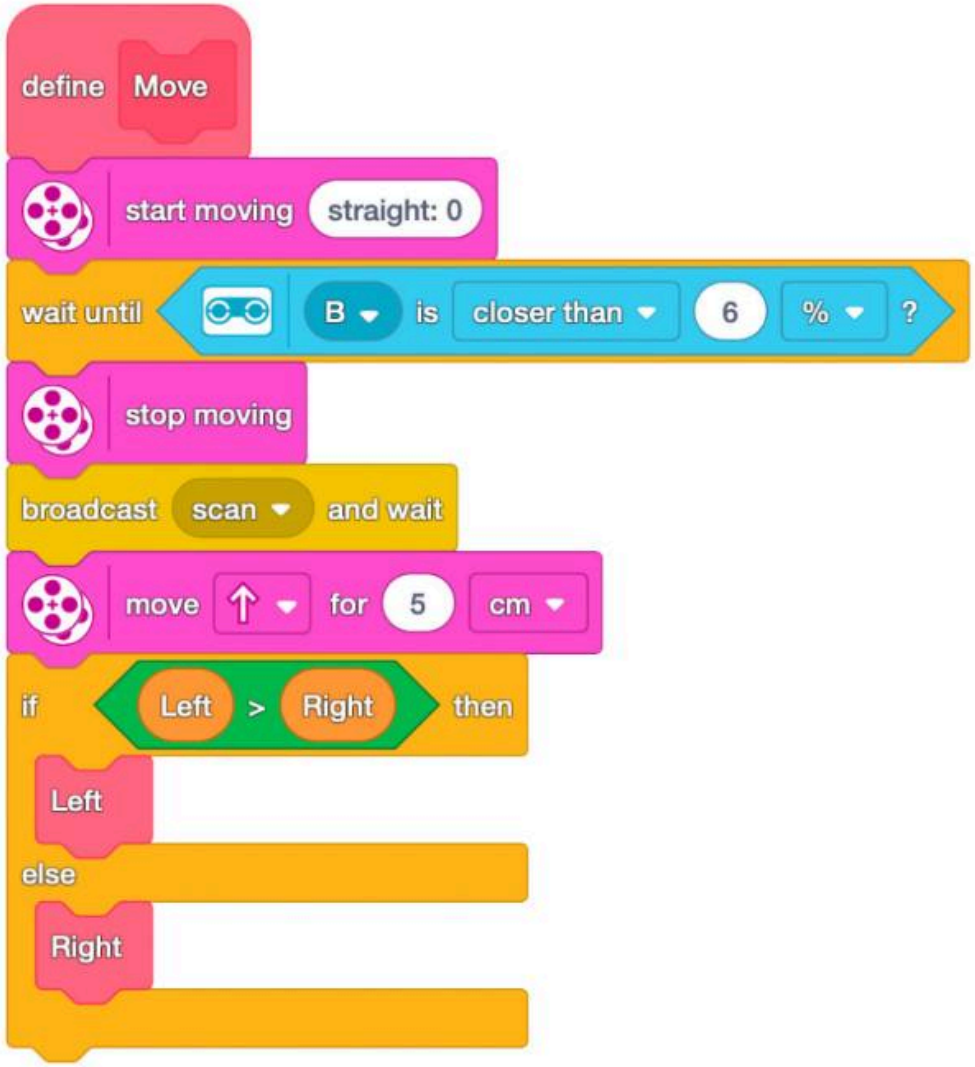
The task performance algorithm can be as follows:

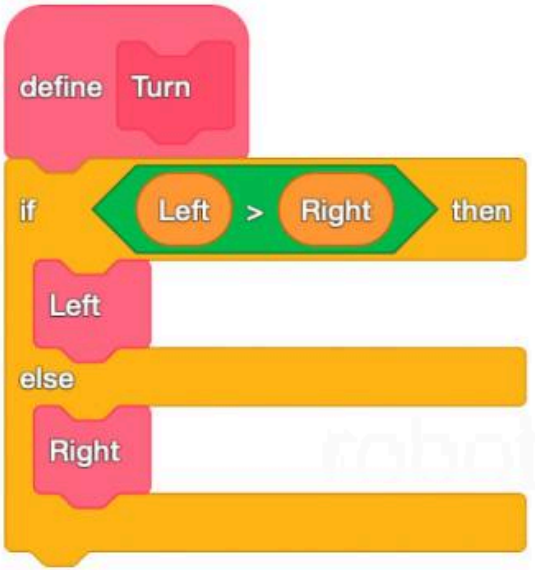


The algorithm for moving to an obstacle should be as follows:



Create the following routines:





The robot turns in the direction in which a longer drive was found

Discuss!

- What is Egypt famous for?
- What is the structure of the Pyramid of Cheops?
- What sensors were used in the robot and for what?
- Why did the robot turn the sensor to the side when turning?
- How many spacewarp generator items have you collected?

